JSS Mahavidyapeetha

JSS SCIENCE AND TECHNOLOGY UNIVERSITY

A Constituent College of JSS STU

JSS Technical Institutions' Campus, Mysuru – 570006

"MODELING AND ANALYSIS OF BIRD WING MORPHING AIRCRAFT"

Thesis submitted in partial fulfillment of curriculum prescribed for the award of the degree of

BACHELOR OF ENGINEERING
IN
MECHANICAL ENGINEERING
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CERTIFICATE

This is to certify that the work entitled "MODELING AND ANALYSIS OF BIRD WING MORPHING AIRCRAFT" is a Bonafide work carried out by

_____, _____, _____and _____in Partial fulfillment for the award of the Degree of BACHELOR OF ENGINEERING IN Mechanical ENGINEERING of Sri Jayachamarajendra College of Engineering, JSS Science and Technology University, Mysuru, during the year 2026. It is certified that all corrections/suggestions indicated during CIE have been incorporated into the report. The project report has been approved as it satisfies the academic requirements concerning the project work prescribed for the Bachelor of Engineering degree

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ABSTRACT

This project focuses on the study and analysis of bird wing morphing and its application in aircraft design. Birds are known for their highly efficient and flexible flight capabilities, which are achieved by continuously changing the shape of their wings. They can extend, fold, twist, and adjust their wing surface depending on flight conditions such as gliding, high-speed flight, or landing. This natural ability of birds has inspired engineers to develop morphing wing technology for modern aircraft.

In conventional aircraft, wings are generally rigid and fixed in shape. Although components such as flaps and slats are used to improve performance, they provide only limited adaptability and often increase mechanical complexity and weight. These limitations have led to the development of morphing wings, which can smoothly change their shape during flight to improve aerodynamic performance.

Morphing wings involve different types of shape changes such as span morphing, camber morphing, twist morphing, and sweep morphing. Each of these plays an important role in controlling lift, drag, and stability of the aircraft. By adjusting the wing shape according to flight conditions, it is possible to achieve better efficiency and performance compared to fixed-wing systems.

This project aims to study the basic concepts of bird-inspired morphing wings and understand their working principles. It includes an analysis of aerodynamic factors such as lift, drag, lift-to-drag ratio, and boundary layer behaviour. These factors are essential in determining the efficiency of an aircraft and how morphing can improve performance.

The study also focuses on the materials and mechanisms used in morphing wing structures. Lightweight composite materials such as carbon fiber and glass fiber are commonly used for structural components, while smart materials like Shape Memory Alloys (SMA) are used as actuators to achieve morphing. Flexible skins are also used to allow smooth deformation of the wing surface without affecting aerodynamic performance.

The expected outcome of this project is to demonstrate that morphing wings can significantly improve aerodynamic efficiency by increasing lift, reducing drag, and enhancing overall flight performance. Morphing also improves stability and manoeuvrability, making aircraft more adaptable to different flight conditions.

Overall, this study highlights that bird-inspired morphing wing technology is a promising approach for improving the performance of modern aircraft. It has great potential in applications such as unmanned aerial vehicles (UAVs), surveillance systems, environmental monitoring, and future aerospace developments.

Acknowledgement

We would like to express our sincere and heartfelt gratitude to our project guide for their continuous support, valuable guidance, and constant encouragement throughout the completion of this project. Their expert advice, timely suggestions, and constructive feedback have played a crucial role in shaping this work. Without their guidance, it would have been difficult for us to successfully complete this project.

We also extend our sincere thanks to all the faculty members of the Department of Mechanical Engineering for their support and for sharing their valuable knowledge during our academic journey. Their teaching has helped us build a strong foundation in mechanical engineering, which was essential for carrying out this project.

We would like to express our deep sense of gratitude to JSS Science and Technology University, Mysuru, for providing us with the opportunity and necessary facilities to complete this project. The academic environment and institutional support have greatly contributed to the successful completion of our work.

We would also like to thank our friends and classmates for their cooperation, helpful discussions, and continuous encouragement during the preparation of this report. Their support made the process easier and more effective.

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We would like to acknowledge the researchers and authors whose work has been referred to in this project. Their contributions in the field of morphing wings and aerospace engineering provided valuable insights and helped us gain a deeper understanding of the subject.

We also thank the laboratory staff and technical assistants for their support and assistance whenever required. Their help in providing necessary information and guidance is highly appreciated.

This project has helped us improve our technical knowledge, analytical skills, and understanding of real-world engineering applications. It has also enhanced our teamwork, coordination, and problem-solving abilities.

Finally, we express our deepest gratitude to our parents and family members for their unconditional support, motivation, and encouragement throughout our academic journey. Their constant support has been a source of strength in completing this project successfully.

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1. INTRODUCTION

Birds have been flying for millions of years and have developed wings perfectly suited for flight. One of the most amazing features of bird wings is that they can change shape. A bird can fold its wings, extend them, twist them, and change the surface area — all depending on what it wants to do. This is why birds can hover, dive at high speed, or glide for long distances.

Modern aircraft have rigid, fixed wings. While this works well, it means the aircraft only performs well in one flight condition. Engineers have tried to solve this using flaps and slats — moveable parts that are complicated, heavy, and still far less flexible than bird wings.

This is why researchers now study morphing wings — wings that smoothly change shape during flight, just like bird wings. This is made possible by advances in smart materials and flexible structures.

1.1 What is Wing Morphing?

Wing morphing means changing the shape of a wing during flight. There are four main types of morphing:

- Span morphing: Making the wing longer or shorter
- Camber morphing: Changing the curve of the wing cross-section
- Twist morphing: Twisting the wing along its length
- Sweep morphing: Changing the angle at which the wing sweeps backward

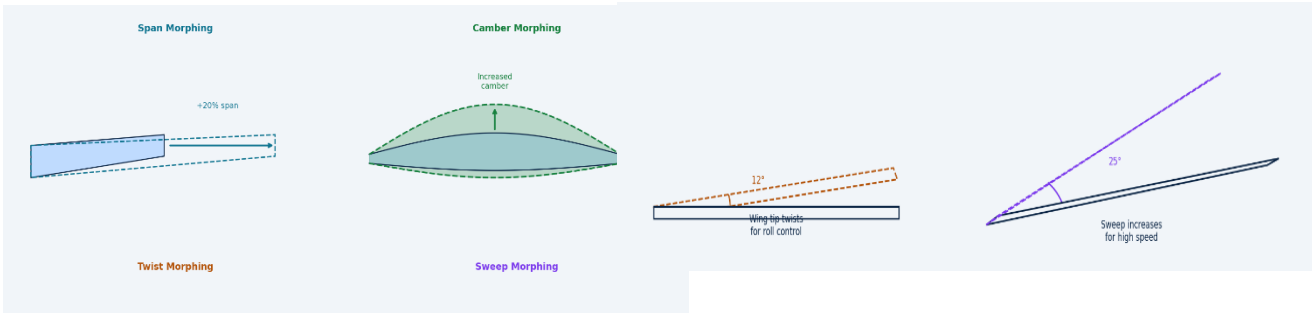


Fig. 1.1: Types of Wing Morphing

2. LITERATURE SURVEY

Many researchers have studied bird wing morphing and tried to apply it to aircraft design. Below is a summary of key studies in this area.

2.1 Early Studies on Bird Flight

The idea of copying bird flight is not new. The Wright brothers studied birds and used wing twisting (wing-warping) in early aircraft design. Later studies showed that birds adjust wing sweep to control speed and lift—sweeping wings back at high speed and spreading them at low speed.

2.2 Shape Memory Alloys in Morphing Wings

Shape Memory Alloys (SMAs) are a popular material for morphing wings. They change shape when heated and return to their original shape when cooled, making them useful as actuators. Solfa et al. (2010) designed a morphing wing using SMA wires to control camber and showed a 15% improvement in lift-to-drag ratio over a fixed wing.

2.3 Shape Memory Alloys in Morphing Wings

Barbarino et al. (2011) reviewed over 100 research papers and found that CFD simulations consistently showed 10–25% drag reduction for morphing wings. Vasista et al. (2012) studied compliant structures and showed that using a flexible structure instead of rigid joints reduces weight and complexity.

2.4 Bio-Inspired Design from Eagles and Gulls

Tucker (1987) found that gliding birds adjust primary feathers to reduce drag — similar to winglets on modern aircraft. Harvey et al. (2019) used scanning to measure a gliding bird wing in detail

2.5 UAV Applications

Ajaj et al. (2016) reviewed morphing wings in UAVs and found the main benefits are longer flight time, better agility, and quieter operation. Main challenges include the added weight of morphing mechanisms and the complexity of the control system.

3. BIRD WING MORPHING CONCEPTS

To design a morphing aircraft wing, it is important to understand the structure and working of bird wings. Birds have highly flexible wings that can change shape during flight, allowing them to achieve efficient and controlled movement.

3.1 Structure of a Bird Wing

A bird wing consists of bones, muscles, tendons, and feathers working together. The bone structure is similar to a human arm. The main bones include:

- Humerus (upper arm)
- Radius and Ulna (lower arm)
- Carpals and Metacarpals (wrist/hand)
- Phalanges (finger bones supporting feathers)

The pectoral muscles are responsible for the downstroke, which generates most of the lift and thrust during flight. Feathers play a key role in controlling airflow and enabling smooth morphing of the wing.

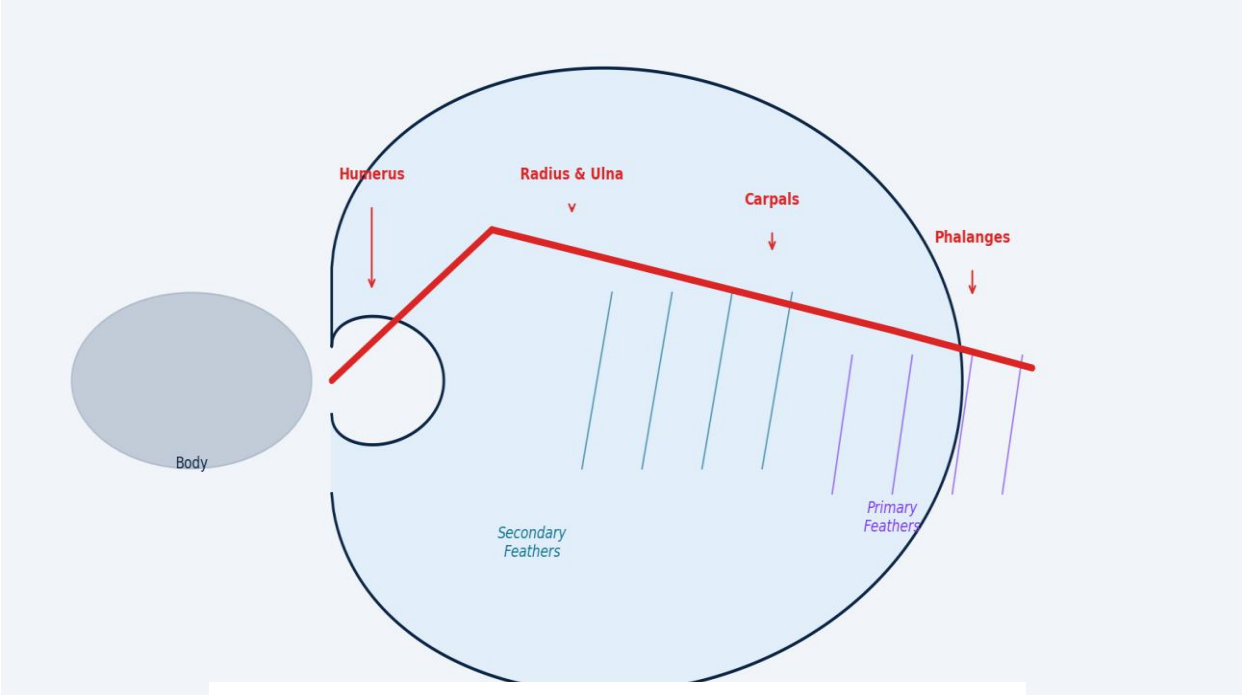


Fig. 3.1: Bird Wing Anatomy

4. AERODYNAMIC PRINCIPLES OF MORPHING WINGS

Aerodynamics is the study of how air flows around objects. For a wing to perform efficiently, the airflow around it must be smooth and controlled. Morphing wings can change their shape during flight, which directly affects lift, drag, and stability, leading to improved aerodynamic performance.

4.1 Lift and Drag

Lift is the upward force that keeps an aircraft in the air. It is generated due to the pressure difference between the upper and lower surfaces of the wing. When the wing has a curved shape (camber), air flows faster over the top surface, reducing pressure and creating lift.

Drag is the force that opposes the motion of the aircraft. It reduces speed and efficiency. By modifying the wing shape, drag can be reduced. For example, reducing camber at high speeds decreases pressure drag, while increasing wingspan helps reduce induced drag.

4.2 The Lift-to-Drag Ratio

The lift-to-drag ratio (L/D ratio) is a key measure of aerodynamic efficiency. A higher L/D ratio means the wing produces more lift for less drag, allowing the aircraft to fly farther with less energy.

For example, albatrosses have an L/D ratio of about 20:1, enabling long-distance gliding, while modern aircraft typically achieve 15–18. Morphing wings aim to improve this ratio across different flight conditions.

4.3 Boundary Layer Control

The boundary layer is a thin layer of air that clings to the wing surface. If it becomes turbulent, drag increases greatly. A laminar (smooth) boundary layer is ideal. Morphing the wing surface can help maintain laminar flow for longer, which reduces drag — one reason why bird feathers are so effective in changing conditions.

5. CONCEPTUAL DESIGN OF THE MORPHING WING

Based on our study of bird wings and aerodynamic principles, we designed a morphing wing for a small UAV with three morphing types: camber change, twist, and span extension.

5.1 Wing Layout

The wing has an aspect ratio of 8 (long and narrow) using a NACA 4412 airfoil as the base shape. It is divided into three sections: an inner fixed zone connected to the fuselage, a middle zone that twists up to 10 degrees, and an outer wingtip that extends up to 20% in span.

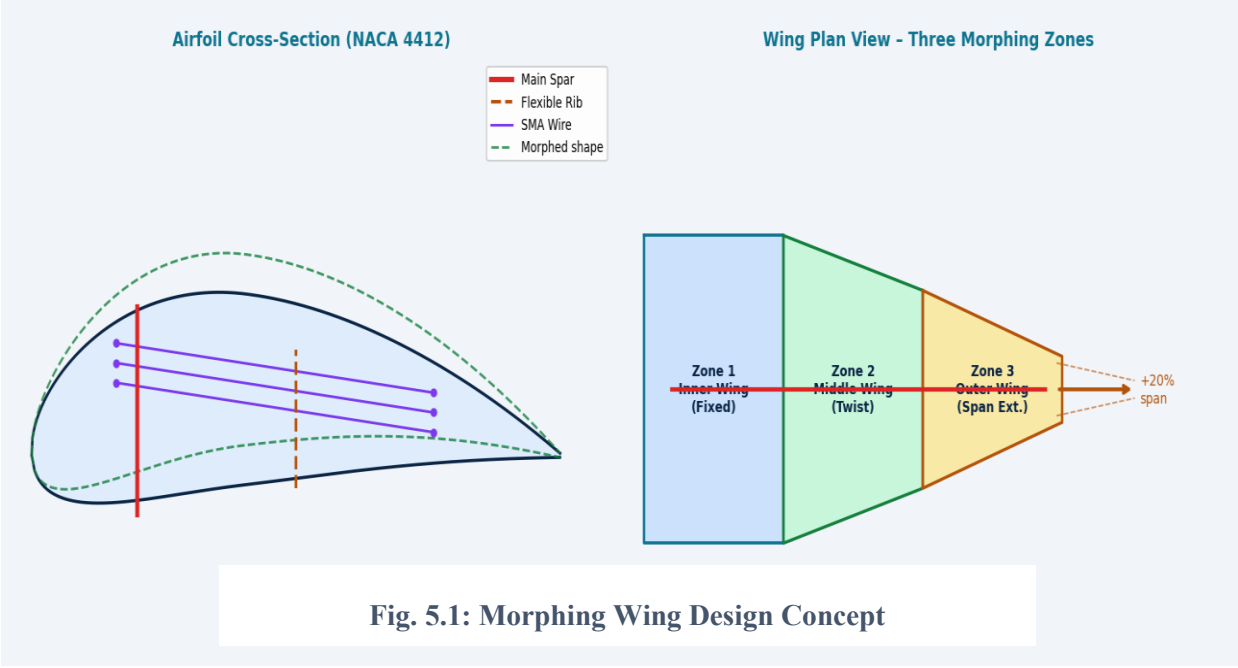


Fig. 5.1: Morphing Wing Design Concept

5.2 Morphing Mechanism

SMA wires embedded in the wing skin contract when heated, pulling the skin to change camber. Flexible composite ribs bend smoothly instead of using rigid joints. A servo motor drives the span extension by pushing a sliding spar to extend the wingtip.

5.3 Comparison of bird Wing-Tail Combinations with the Combinations of Bird Wing morphed UAV

i. Combinations While Landing

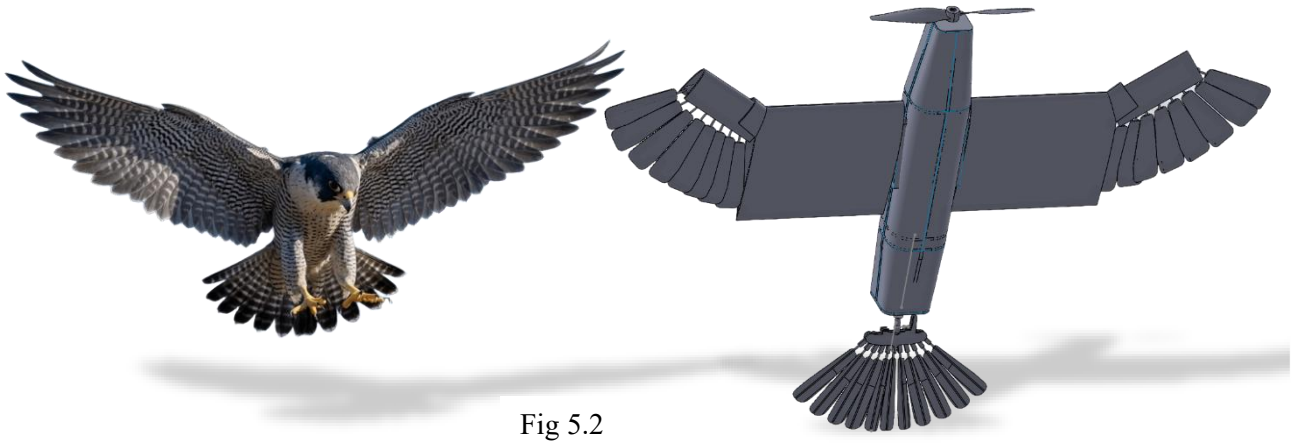


Fig 5.2

ii. Combinations While Diving

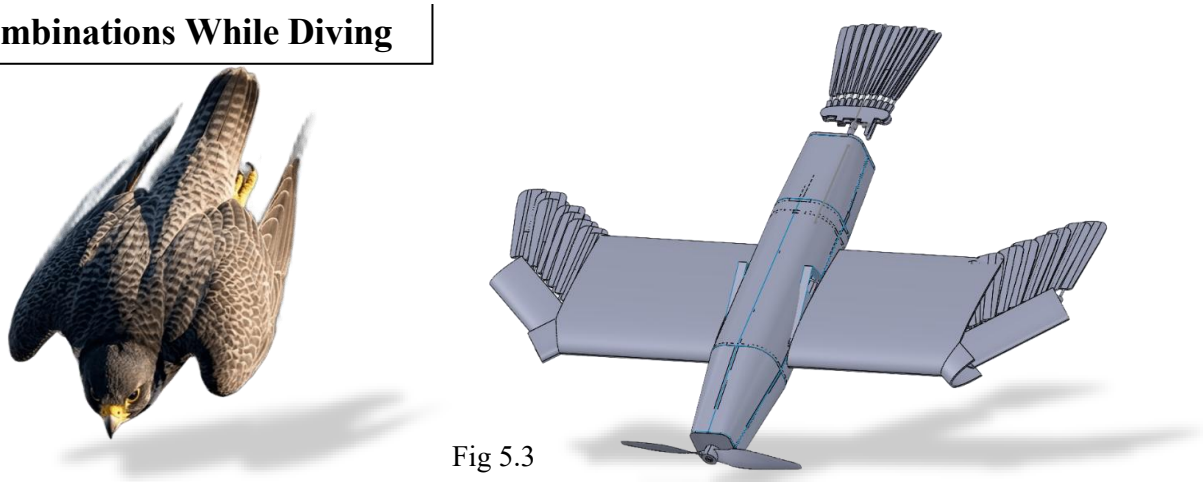


Fig 5.3

iii. Combinations While Diving

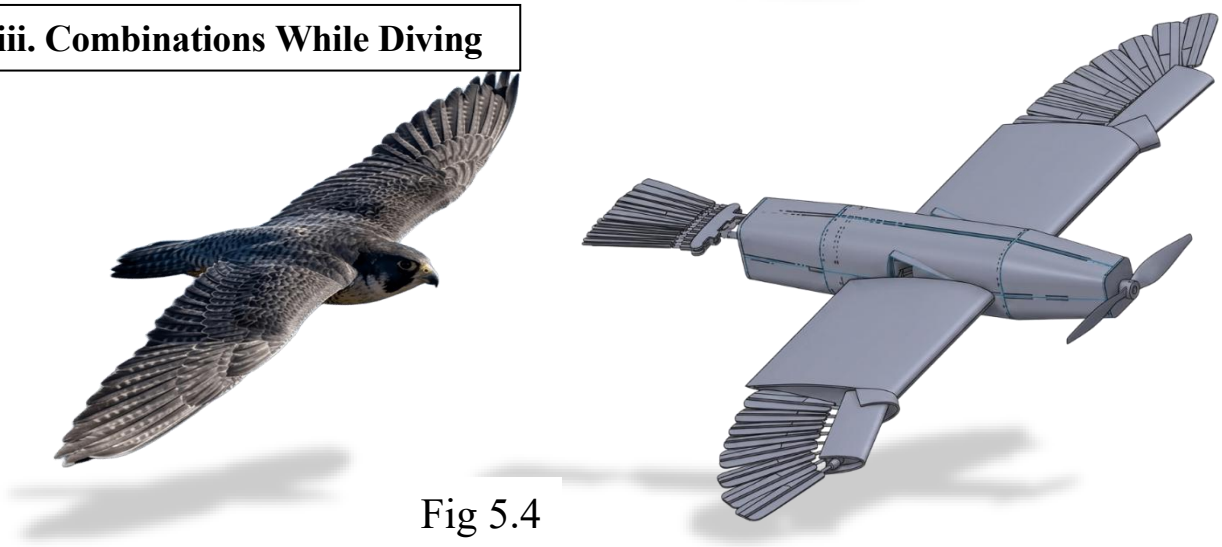


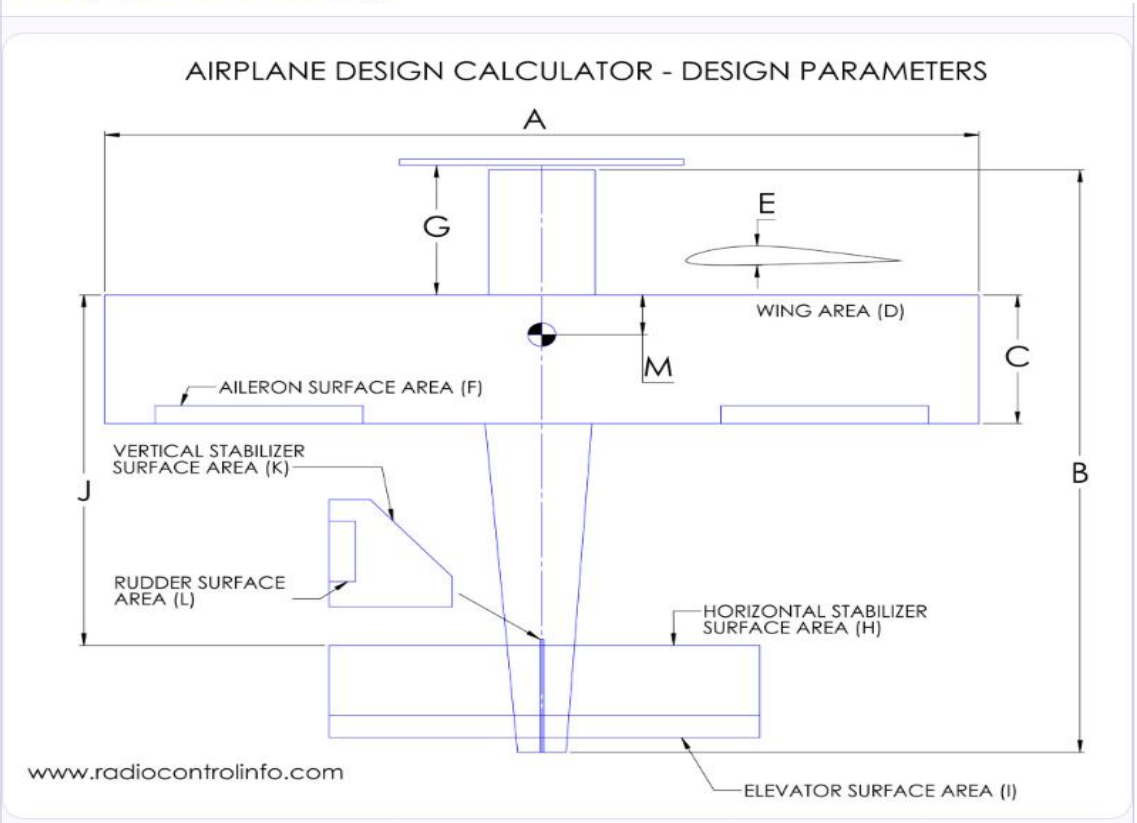
Fig 5.4

6.CALCULATIONS FOR THE RC PLANE MODEL

6.1 Calculations by { https://www.radiocontrolinfo.com/ }

RC Cars, Boats, Planes and Calculators.

▼ Design-parameter reference image



The letters in the results match the diagram, including wingspan A, fuselage length B, wing chord C, wing area D, wing thickness E, aileron area F, prop/nose-to-wing distance G, horizontal stabilizer H, elevator I, tail distance J, vertical stabilizer K, rudder L, and CG range M.

AIRPLANES

RCexplained Airplane Design Calculator

Estimate starting airframe dimensions, control surface areas, target weight, power, and CG range from wingspan and airplane type.

Design workflow: This calculator is intended as a foam-board style starting point. Use the results to sketch a rectangular-wing design, then refine the design based on your structure, airfoil, equipment, and flying goals.

▼ Airframe selection

Setup name

My airplane design

Used when saving this airplane design to your member profile.

Airplane type

Sport

Select the general style of airframe you are designing.

Wingspan, inches

32

Enter the planned wingspan. This calculator is intended for typical RC model wingspans from about 18 to 135 inches.

Calculate Airplane Design | Reset Defaults

Airplane design results

Use these values as a first-pass design target for a simple rectangular-wing foam-board aircraft. Wing loading, power loading, and CG range are the key values to check before building.

DESIGN READ

Sport

Sport starting point: faster and more responsive than the trainer layout. Keep the structure straight and be more conservative with first-flight throws.

Main wing and fuselage

WINGSPAN (A)	FUSELAGE LENGTH (B)	WING CHORD (C)
32.00 in	27.20 in	5.82 in
input wingspan	starting fuselage length	rectangular wing assumption

WING AREA (D)	WING THICKNESS (E)	NOSE/PROP TO WING LE (G)
186 sq in	0.76 in	5.76 in
wingspan × chord	starting airfoil thickness	nose or prop to main wing leading edge

Control surfaces and tail

TOTAL AILERON AREA (F)	AILERON SIZE STARTING POINT	HORIZONTAL STABILIZER AREA (H)
20.5 sq in	8.47 × 1.21 in	46.5 sq in
left + right ailerons	length × width per aileron	including elevator

ELEVATOR AREA (I)	WING LE TO TAIL LE (J)	VERTICAL STABILIZER AREA (K)
11.6 sq in	19.8 in	18.6 sq in
starting elevator surface area	main wing leading edge to tail leading edge	including rudder

RUDDER AREA (L)
4.7 sq in
starting rudder surface area

Weight, power, and CG

RECOMMENDED MAX AUW	RECOMMENDED POWER	WING LOADING
14.8 oz	93 W	11.5 oz/ft²
target all-up weight	starting power target	based on target AUW

POWER LOADING	CENTER OF GRAVITY (M)
100 W/lb	1.28 to 1.75 in
power compared with target weight	22% to 30% of wing chord

Build note: These values are approximations for simple foam-board-style designs with a constant wing chord. Final stability still depends on airfoil shape, structure, control throws, incidence, real finished weight, and careful CG testing.

6.2 Considered dimensions accordingly as per our model

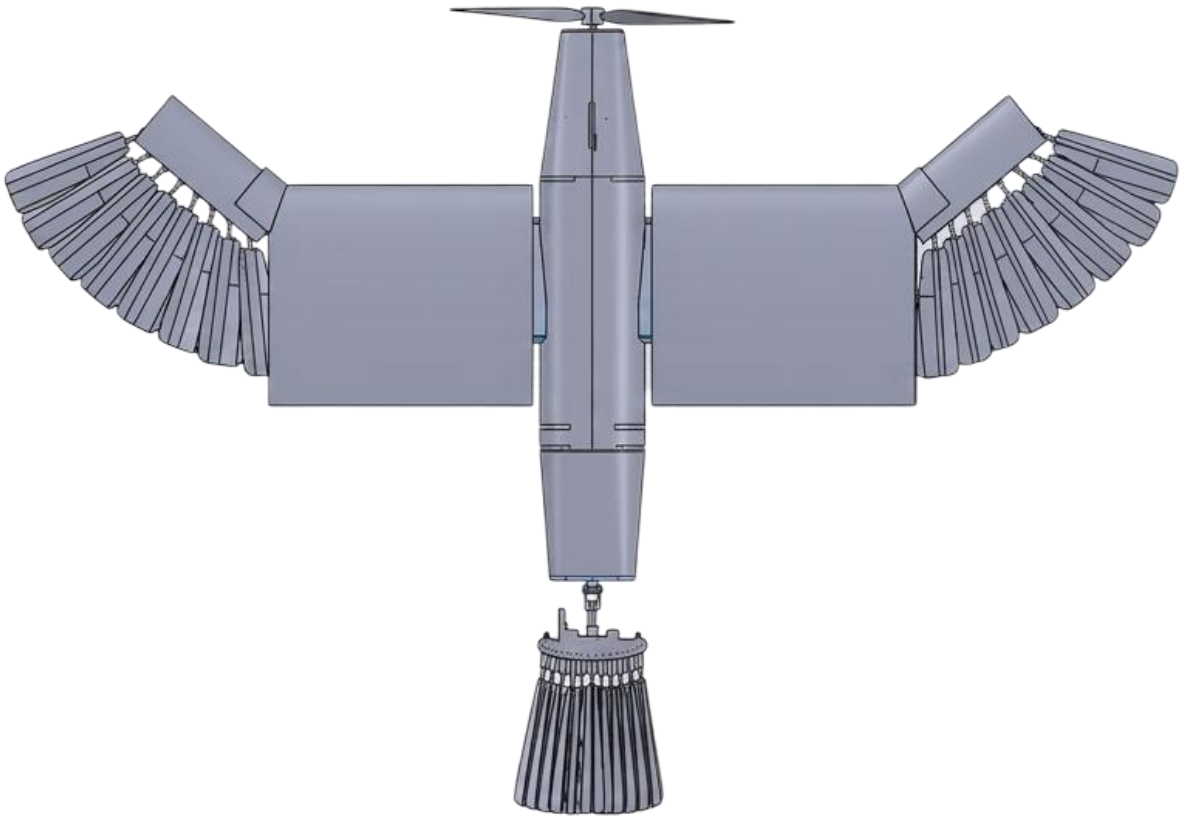
1. **WINGSPAN [A]: 800mm {31.49 ≈ 32 inches}**
2. **FUSELAGE [B]: 450mm {17.7165 ≈ 18 inches}**
3. **WING CHORD [C]: 160mm {6.69291 inches}**
4. **WING AREA [D]: (while wings fully opened as of Fig5.4)**

Wing Area (D) = Wingspan (b) × Chord (c)

= 800*170

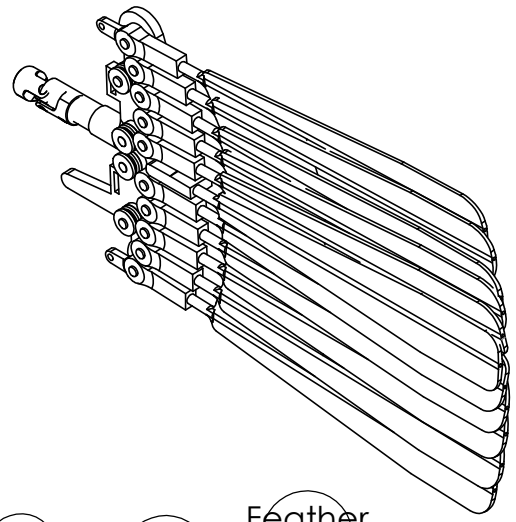
=136000mm^2 = 210.800422 inch^2

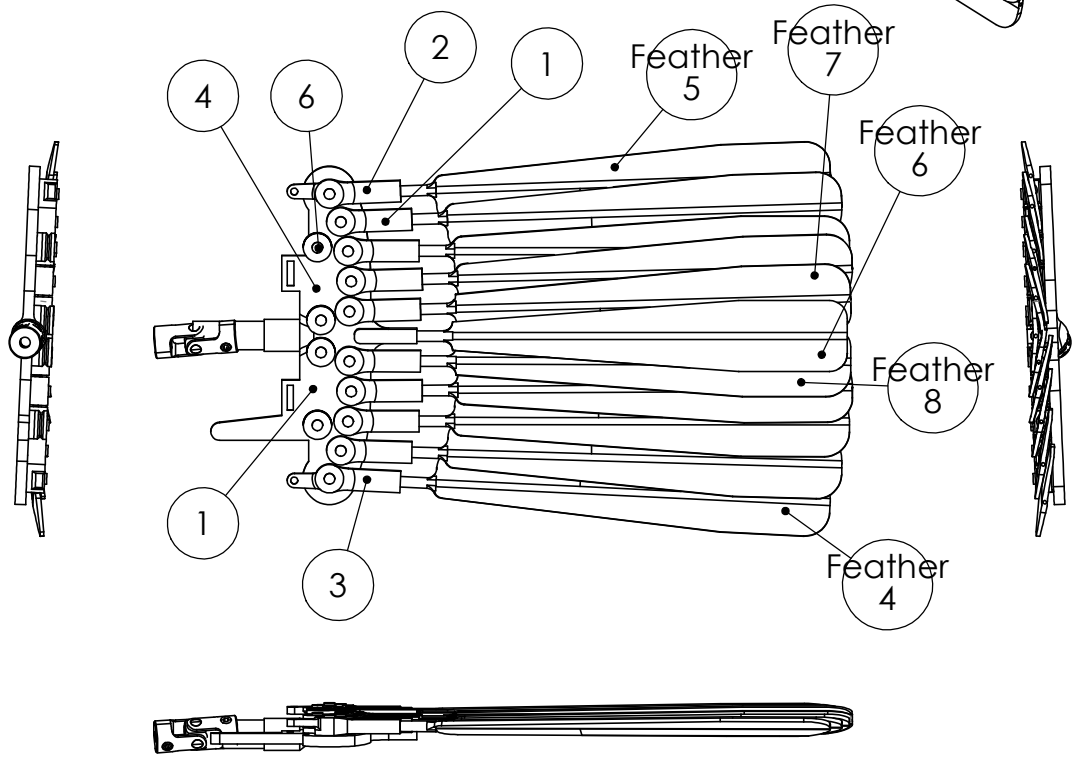
5. **WING THICKNESS [E]: 27mm {1.06299 inches}**
6. **CENTER OF GRAVITY [M]: 50.59mm {1.9917323 inch}**



7. DRAFTS OF THE BIRD WING MORPHED UAV

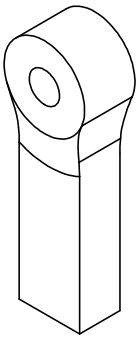
Tail Part Mechanism

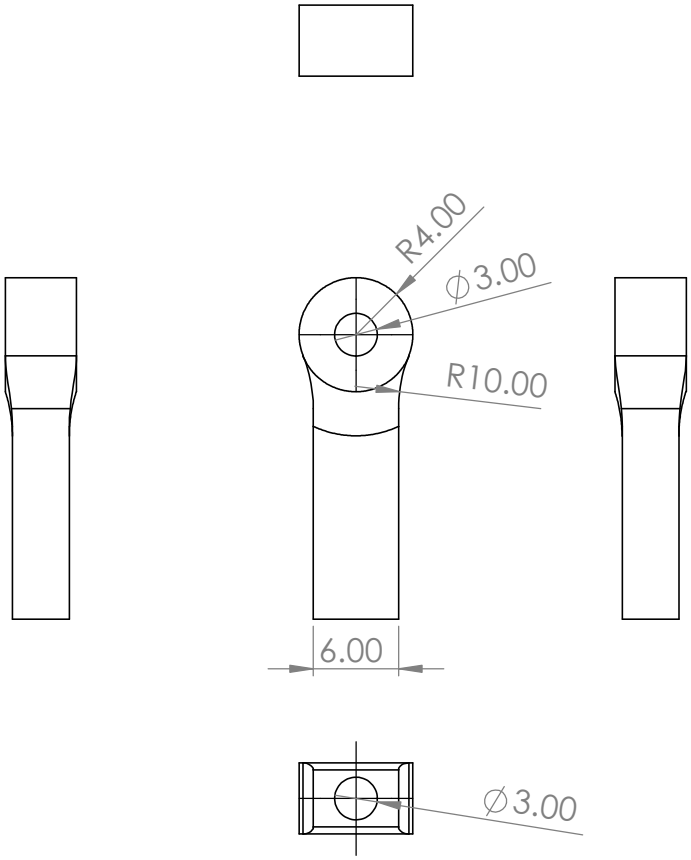




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	NAME	SIGNATURE	DATE				TITLE:	
DRAWN								
CHK'D								
APPV'D								
MFG								
Q.A				MATERIAL:			DWG NO. Tail mechanism	A4
				WEIGHT:			SCALE:1:5	SHEET 1 OF 1

1. Tail Feather Holder





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APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO. feather holder drawing

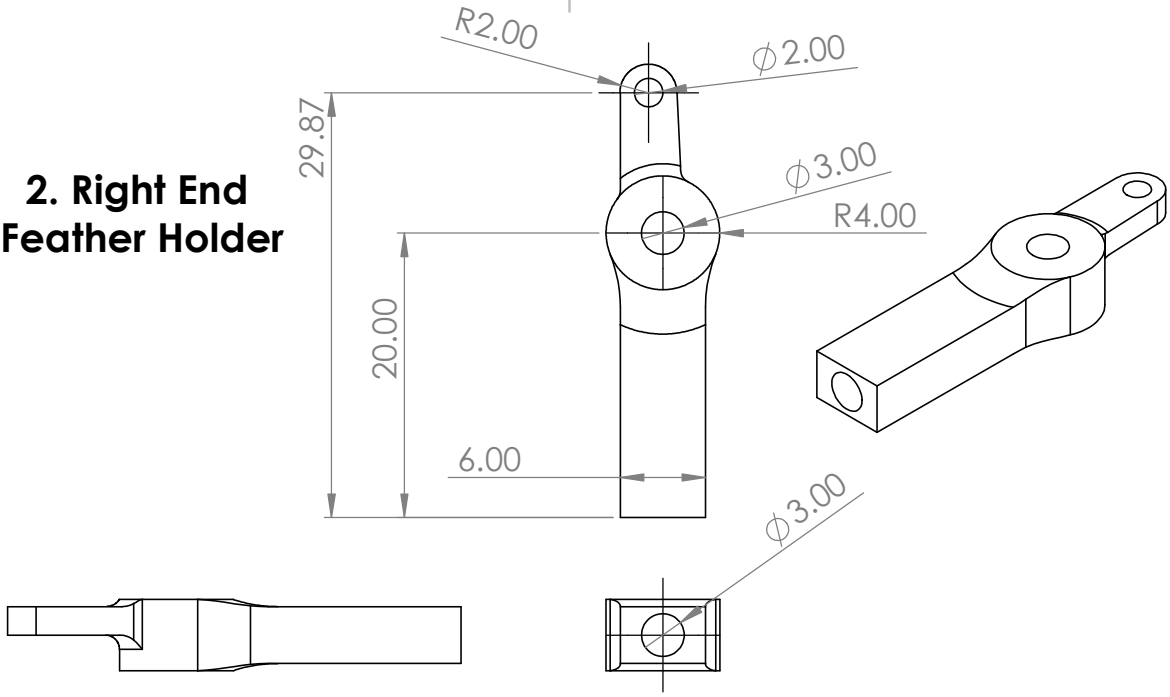
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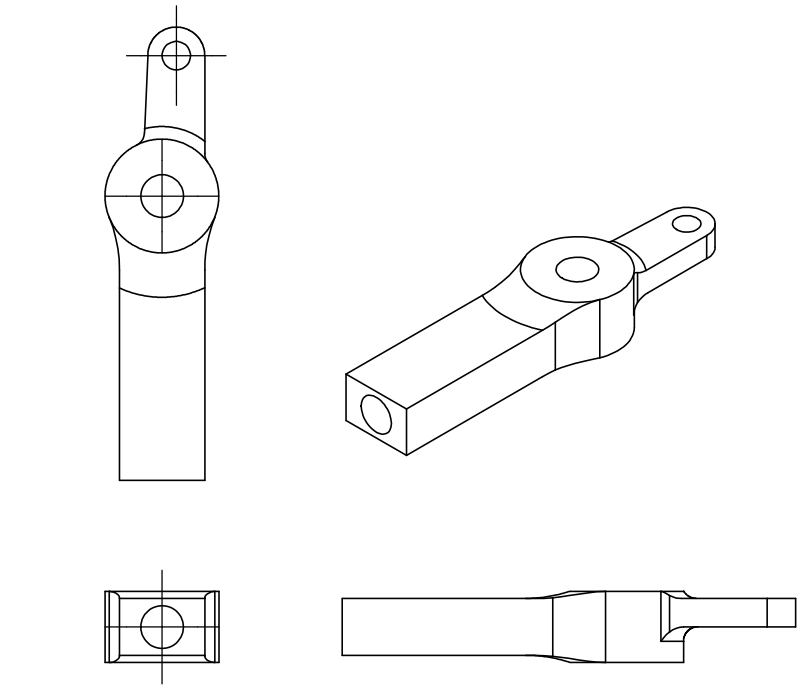
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SHEET 1 OF 1

2. Right End Feather Holder



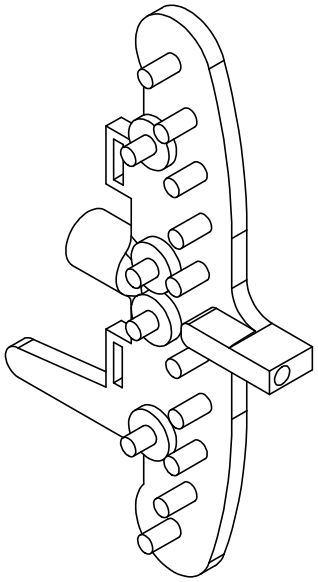
3. Left End Feather Holder

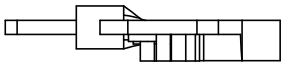


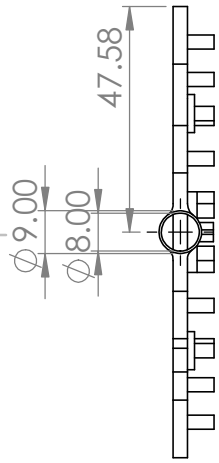
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MFG							
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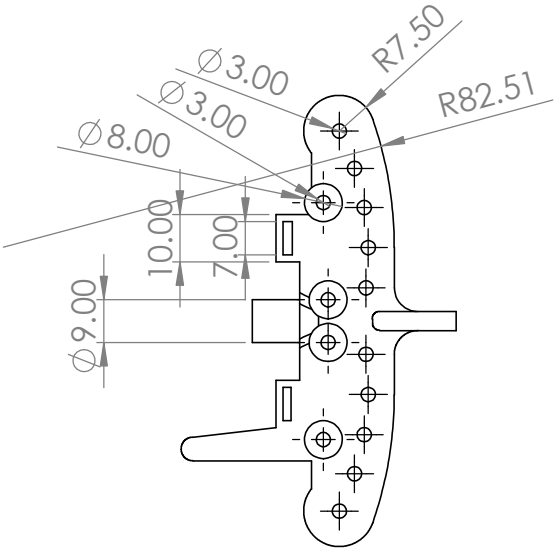
Right and left end feather holder

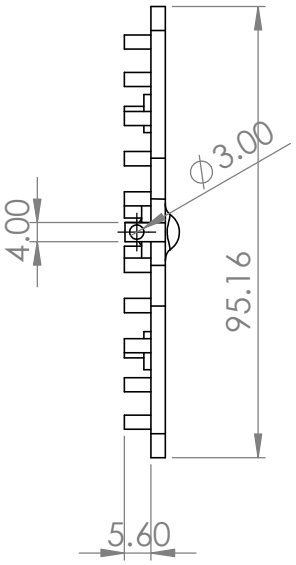
4. Tail Mechanism Base

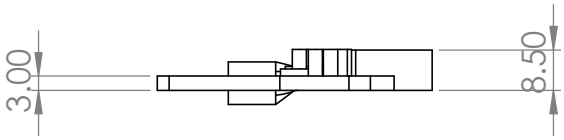












UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:	DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO. Tail Mechanism Base

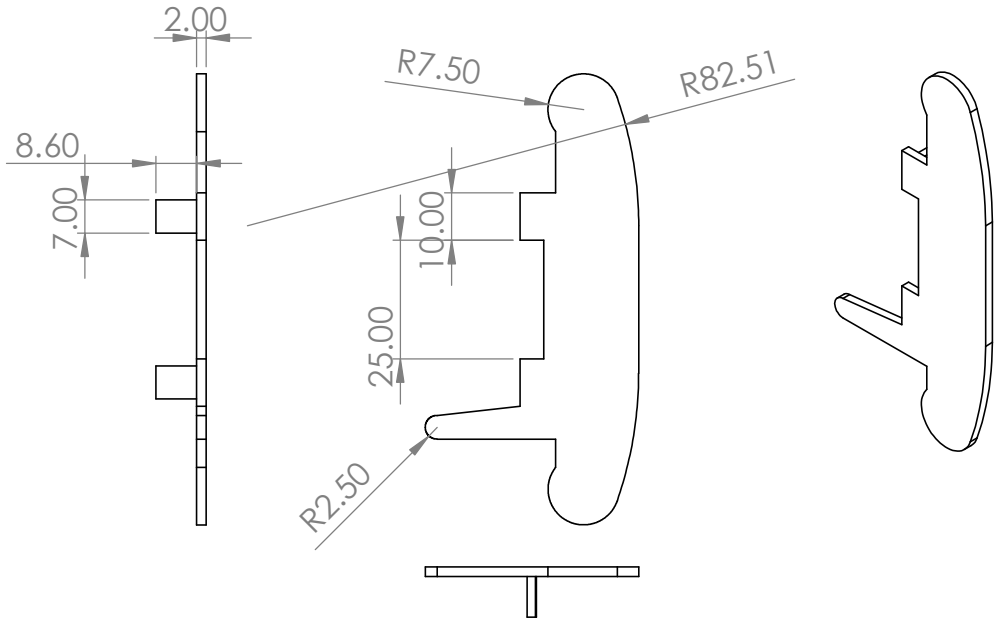
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WEIGHT:

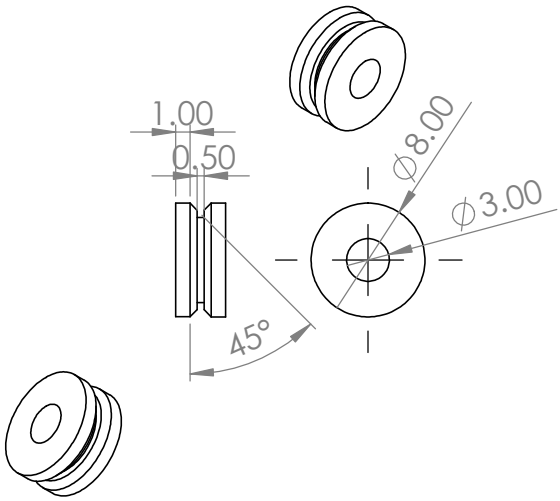
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SHEET 1 OF 1

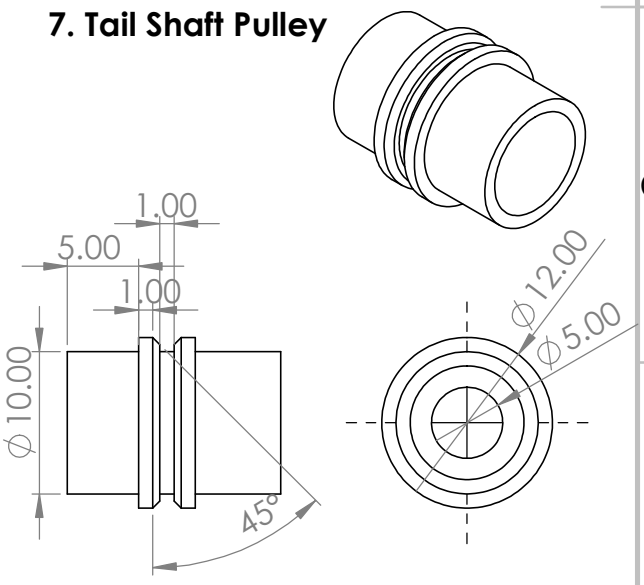
5. Tail Part Mechanism Base Cover



6. 8mm Pulley



7. Tail Shaft Pulley



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:	DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO. Tail Parts

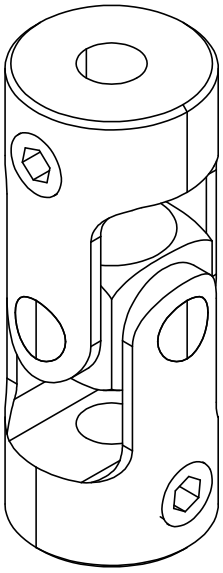
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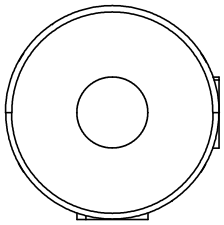
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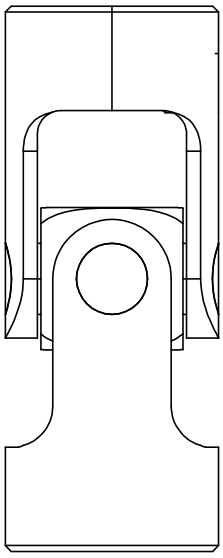
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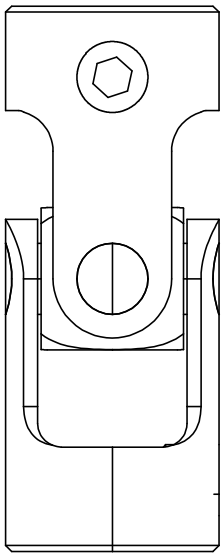
SHEET 1 OF 1

Universal Joint





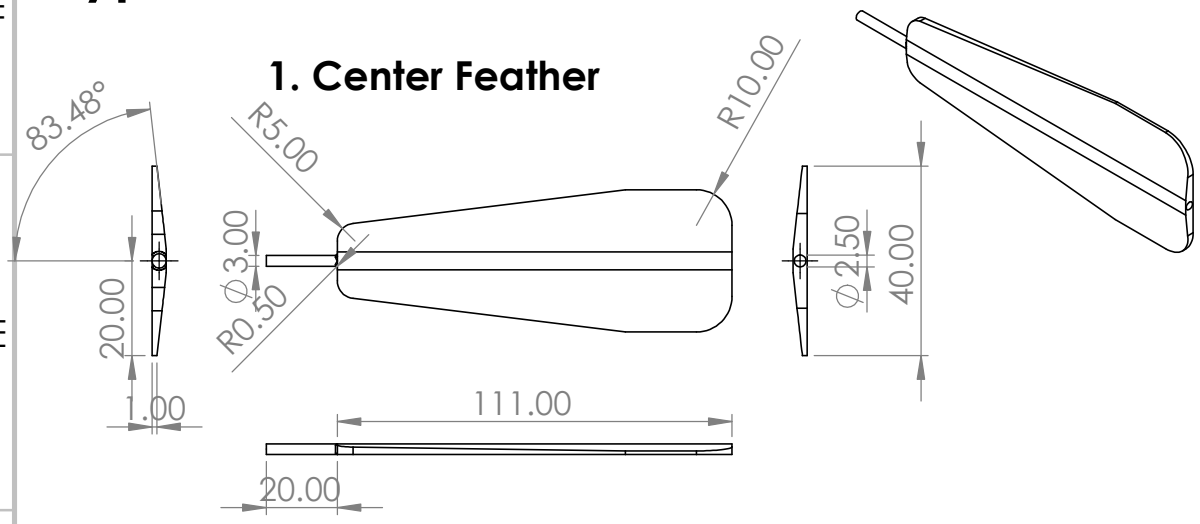




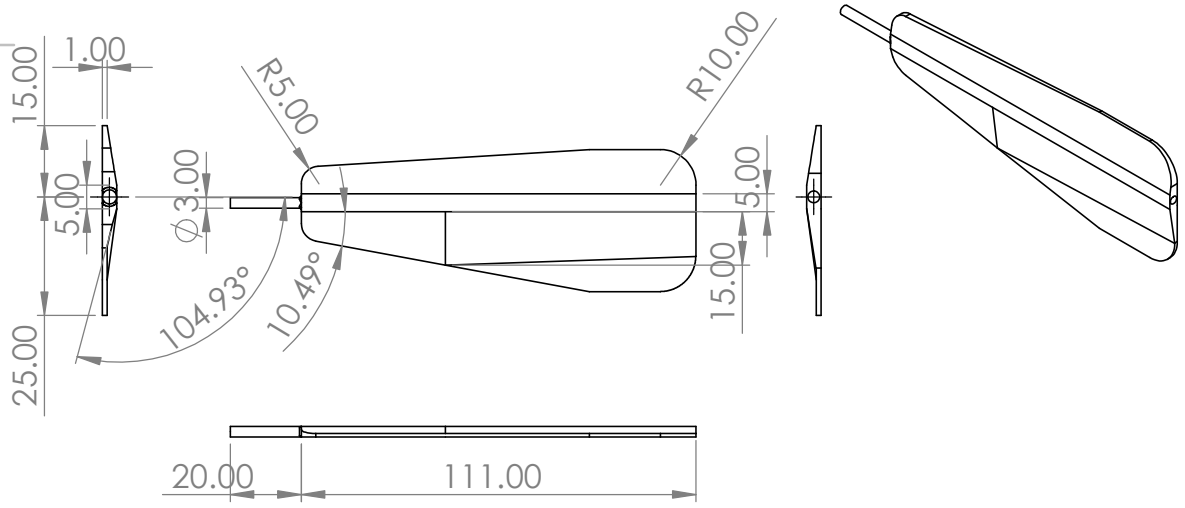
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Universal Joint	A4
				WEIGHT:		SCALE:5:1	SHEET 1 OF 1

Types Of Feathers Used

1. Center Feather

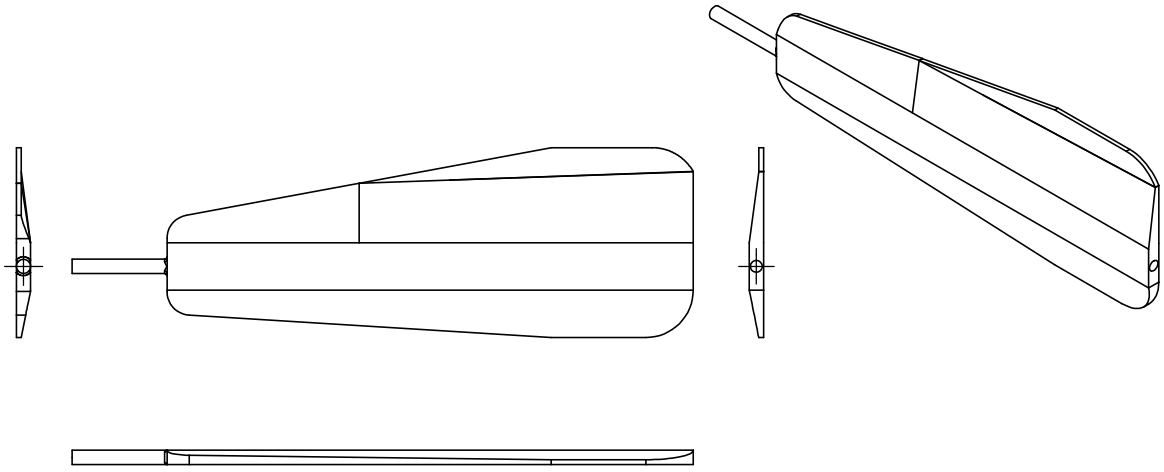


2. Right Side Feathers (Wing Part)

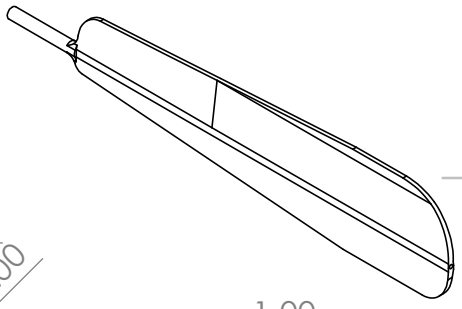


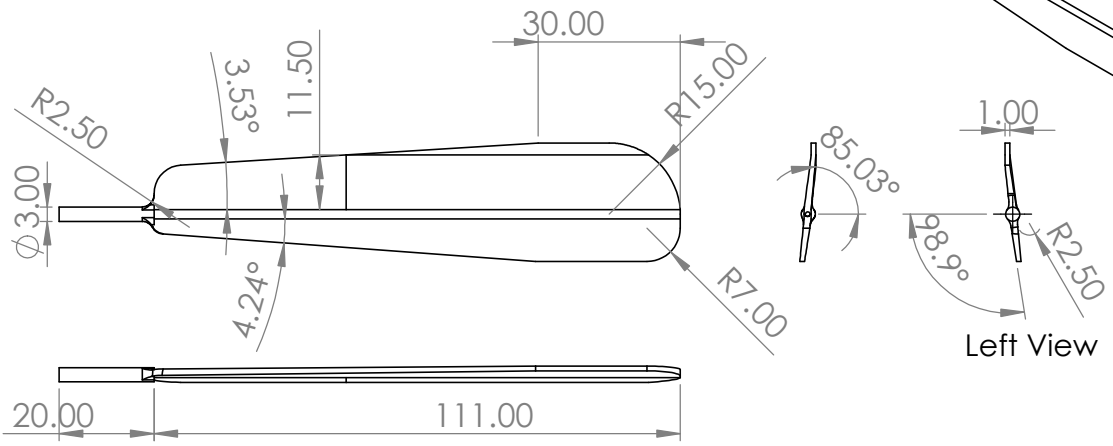
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Feathers types 1	A4
				WEIGHT:		SCALE:1:2	SHEET 1 OF 1

3. Left side Feather (Wing Part)



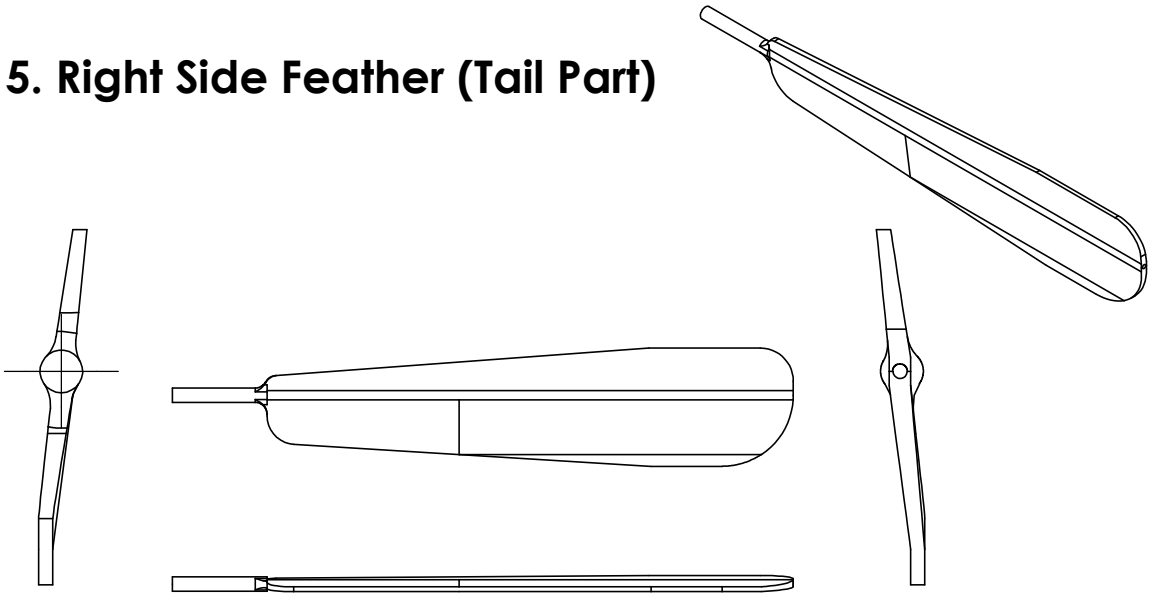
4. Left Side Feather (Tail Part)



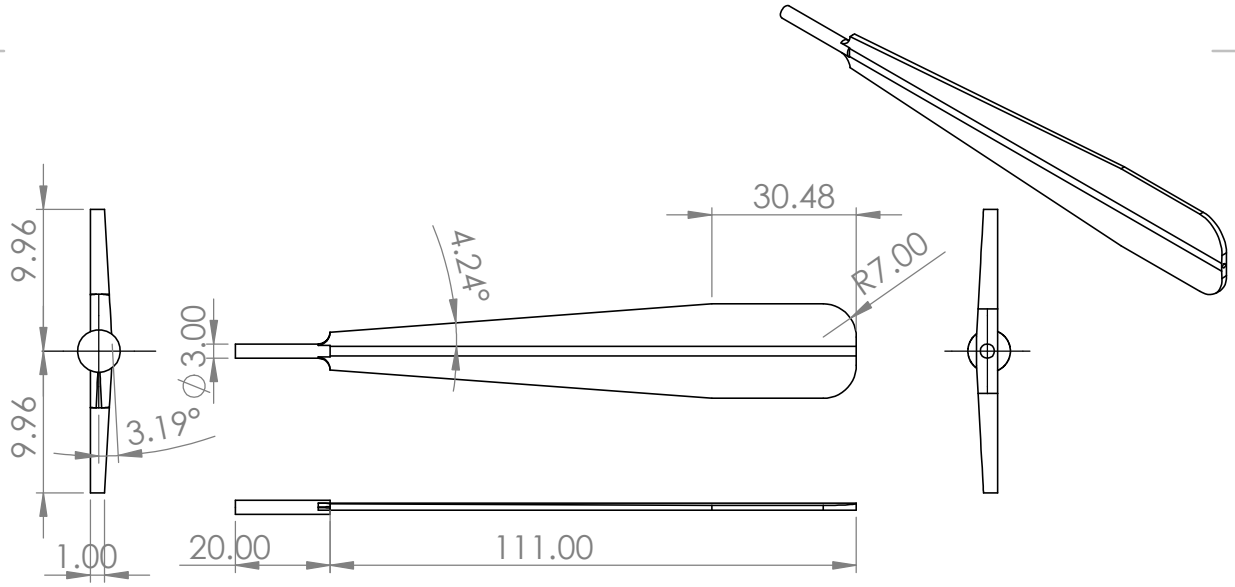


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO.	A4
						Feathers types 2	
				WEIGHT:		SCALE:1:2	SHEET 1 OF 1

5. Right Side Feather (Tail Part)



6. Double Side Feather (Tail Part)



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:	DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE:

MATERIAL:

DWG NO. Feathers types 3

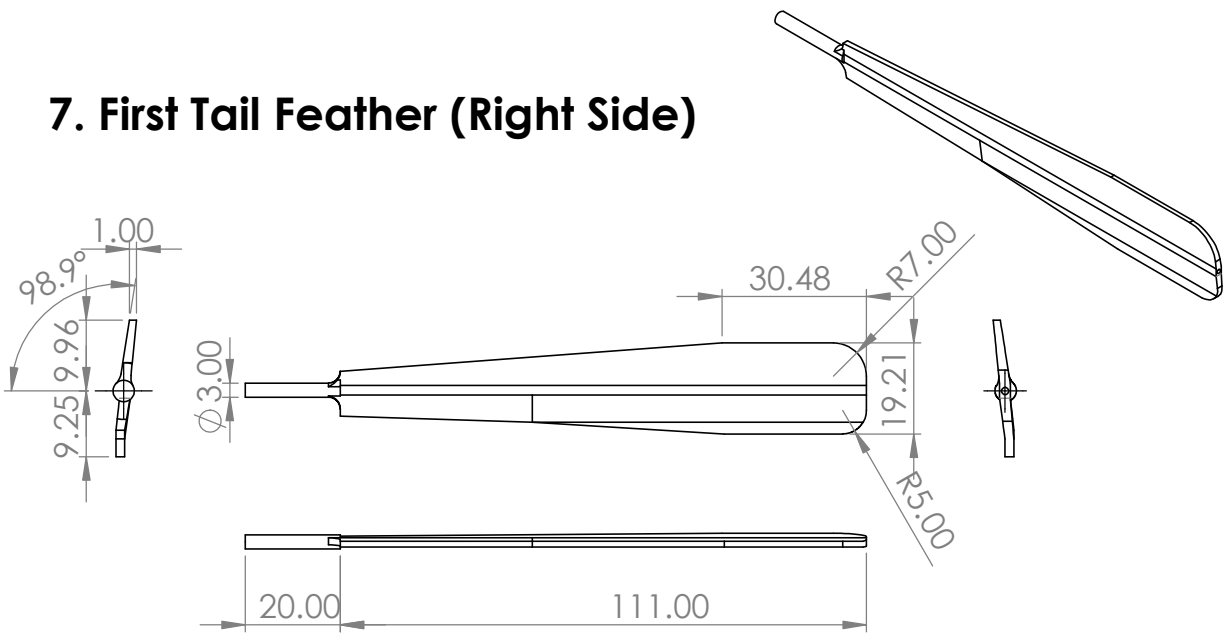
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WEIGHT:

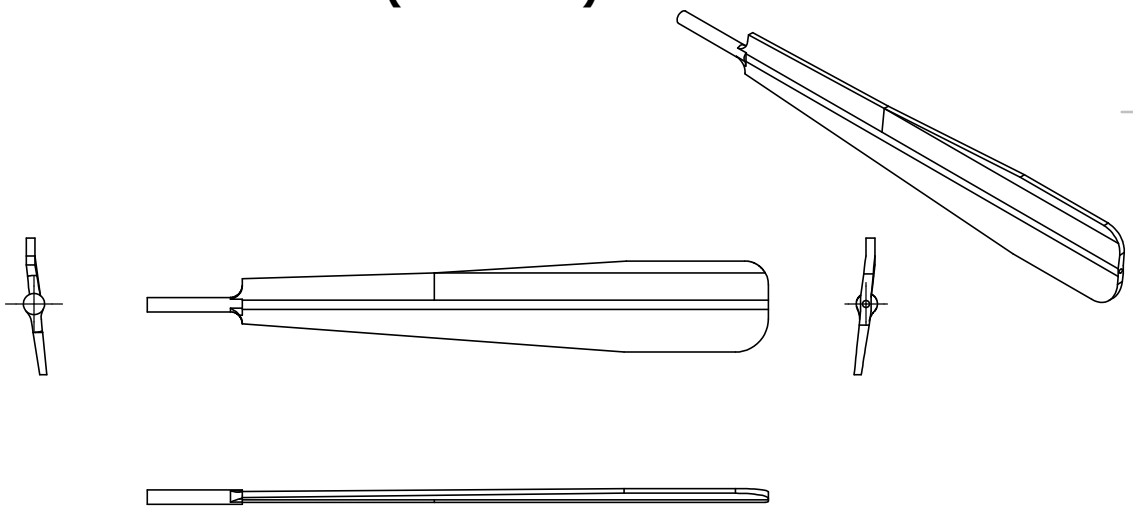
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SHEET 1 OF 1

7. First Tail Feather (Right Side)

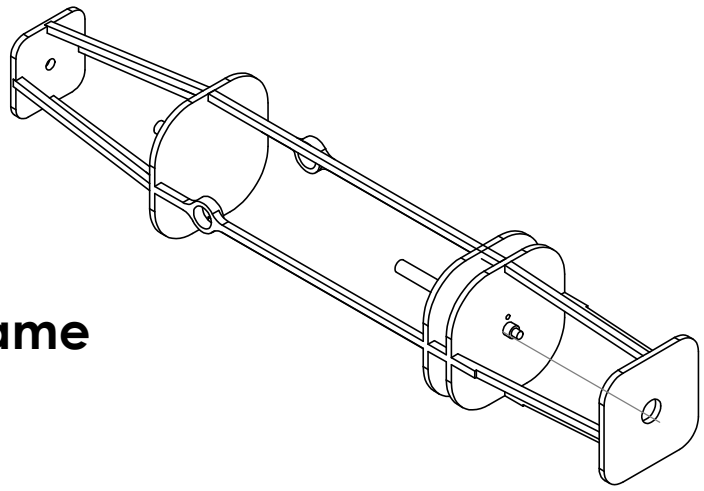


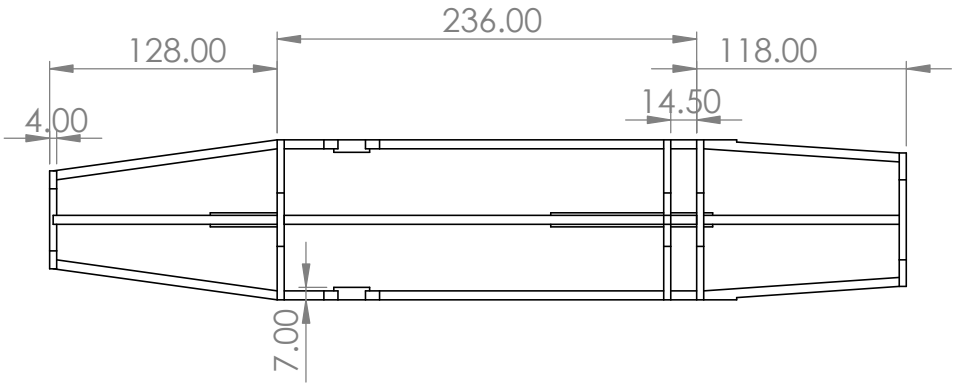
8. First Tail Feather (Left Side)

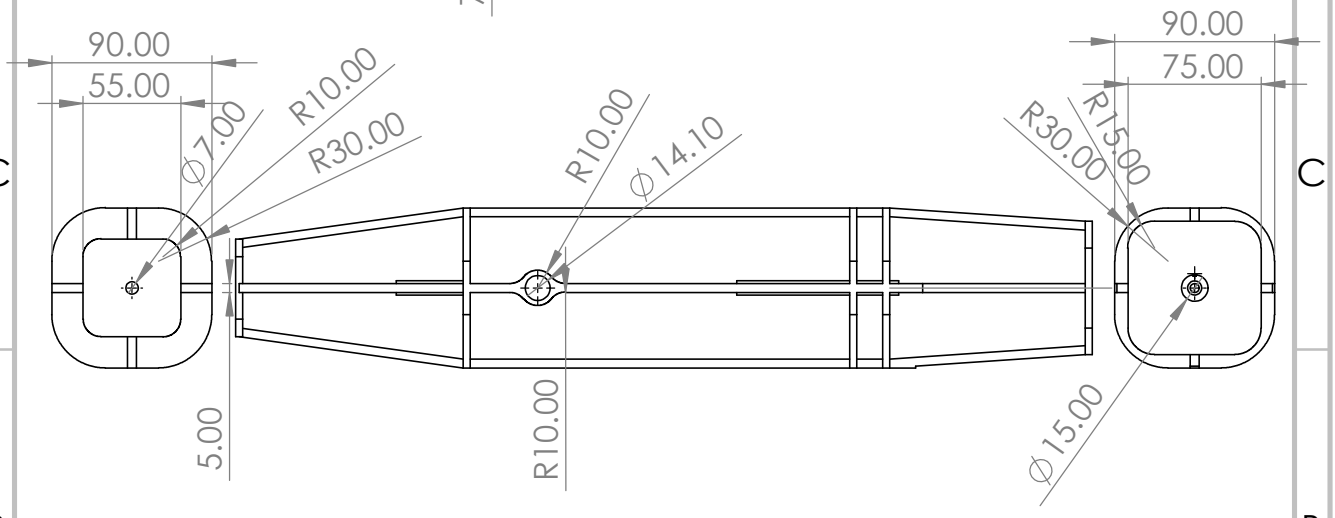


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:	
DRAWN								
CHK'D								
APPV'D								
MFG								
Q.A				MATERIAL:			DWG NO. Feathers types 4	A4
				WEIGHT:			SCALE:1:2	SHEET 1 OF 1

Fuselage body Frame

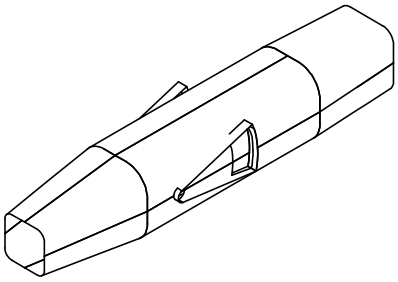


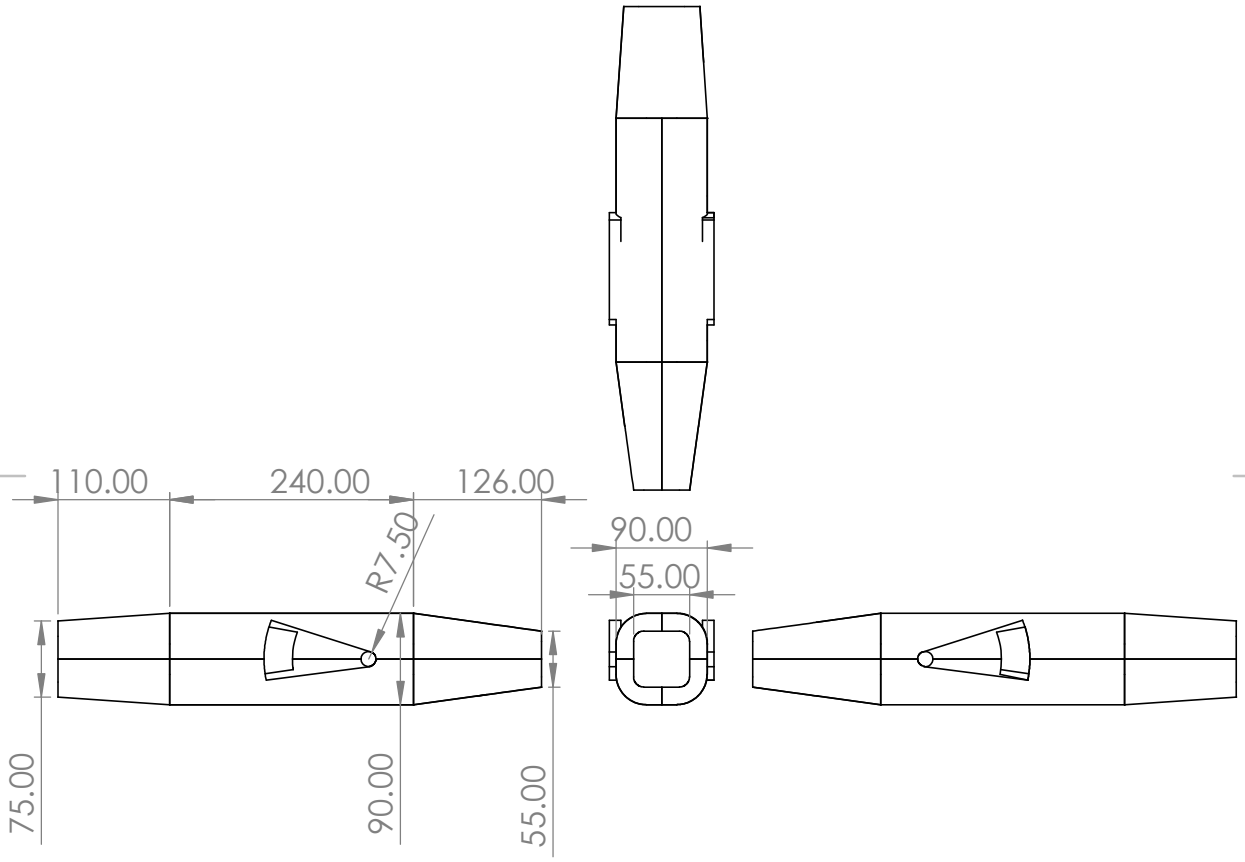




UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:	
DRAWN								
CHK'D								
APPV'D								
MFG								
Q.A				MATERIAL:			DWG NO.	A4
							Fuselage Body Frame	
				WEIGHT:			SCALE:1:5	SHEET 1 OF 1

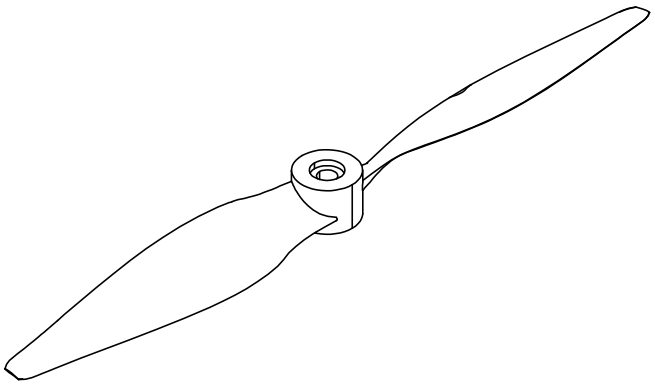
Fuselage Body

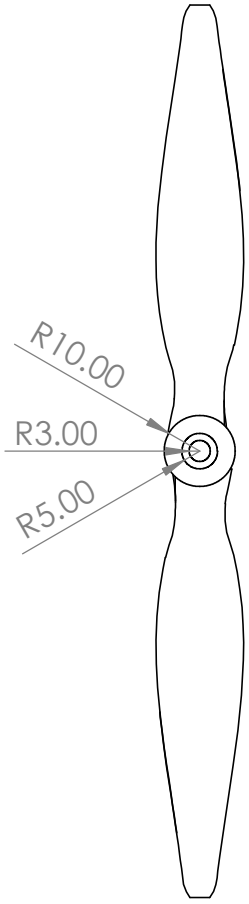


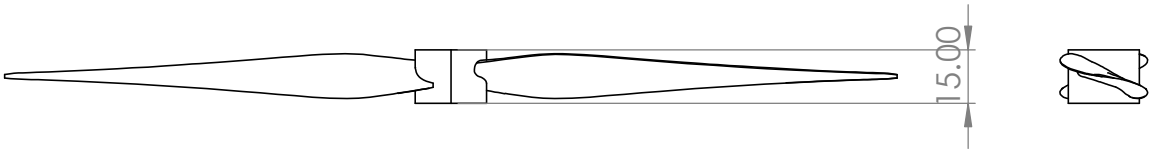


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO. Fuselage Body	A4
				WEIGHT:		SCALE:1:10	SHEET 1 OF 1

Propeller

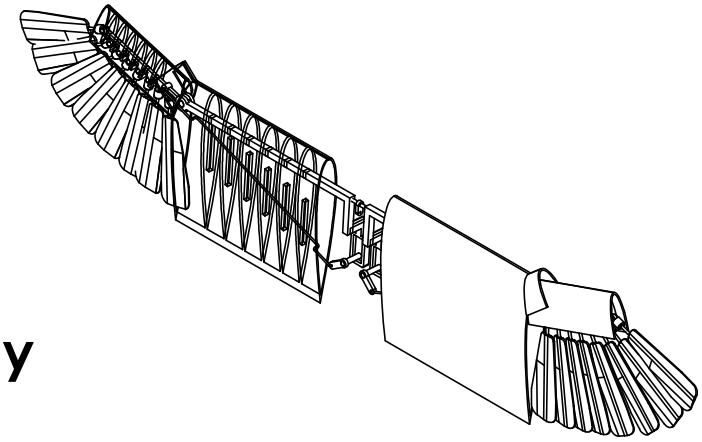


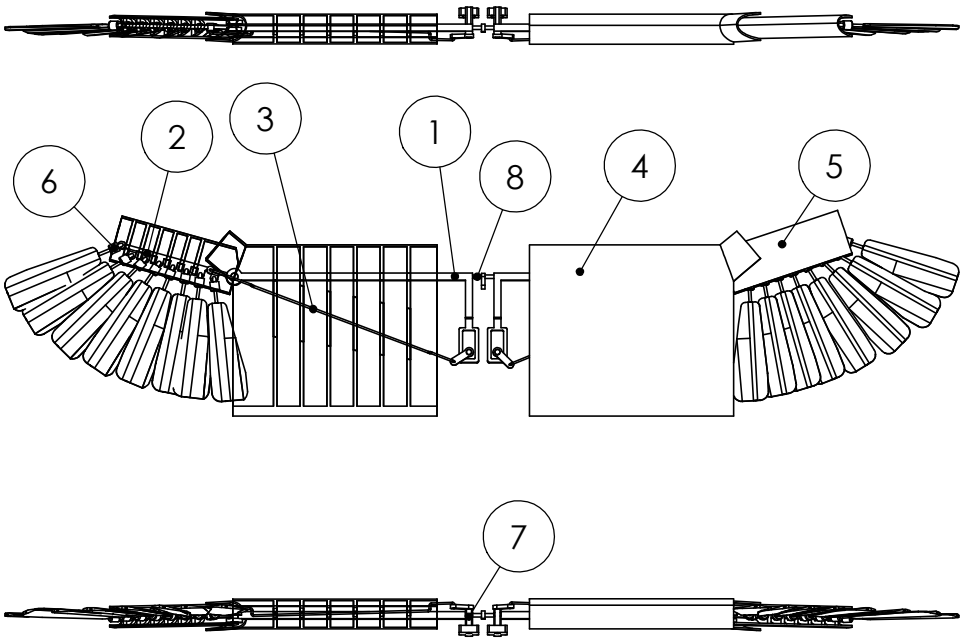




UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION	
	NAME	SIGNATURE	DATE				TITLE:
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO.	A4
						Propeller 10x5.5	
				WEIGHT:		SCALE:1:5	SHEET 1 OF 1

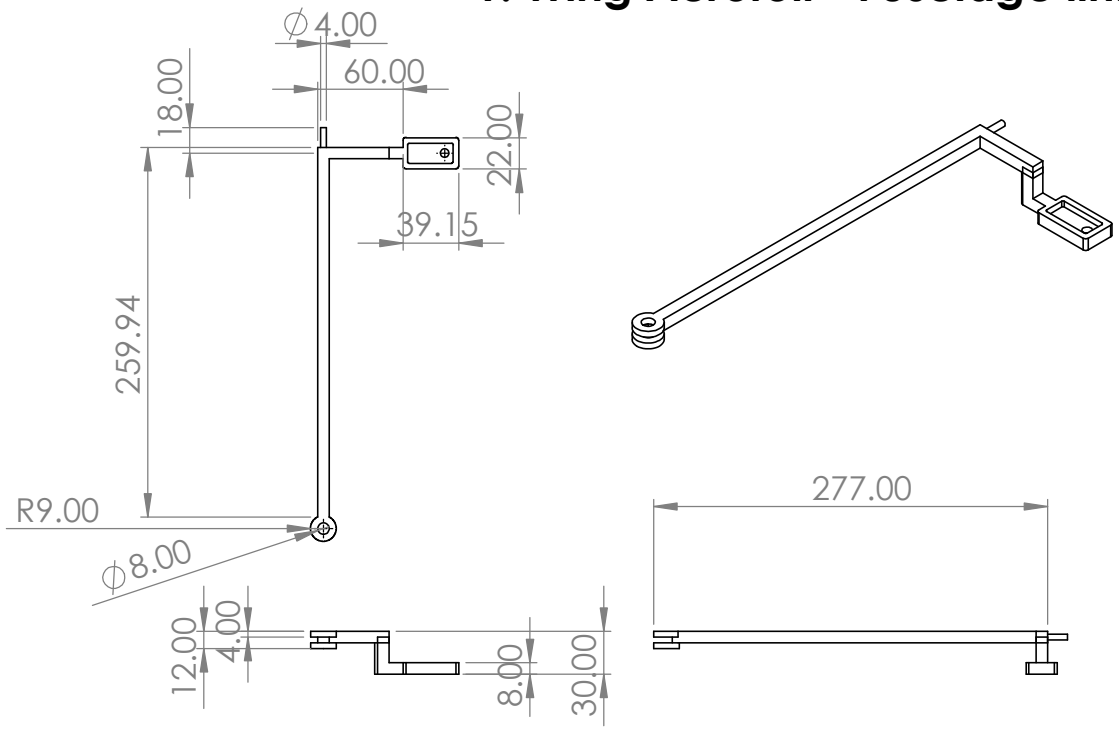
Wings Assembly



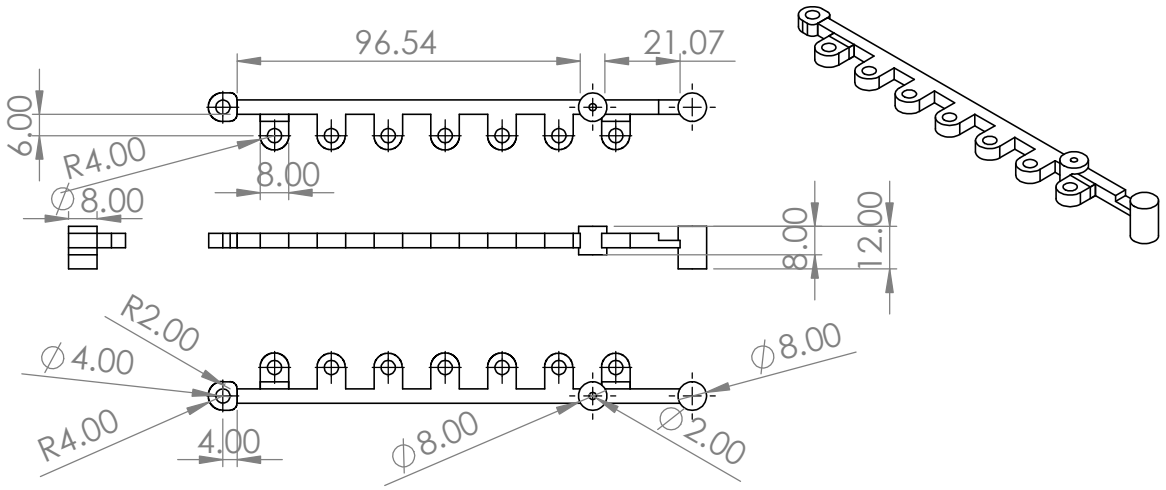


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:	
DRAWN								
CHK'D								
APPV'D								
MFG								
Q.A				MATERIAL:			DWG NO. Wings Assembly	A4
				WEIGHT:			SCALE:1:20	SHEET 1 OF 1

1. Wing Aerofoil - Fuselage link

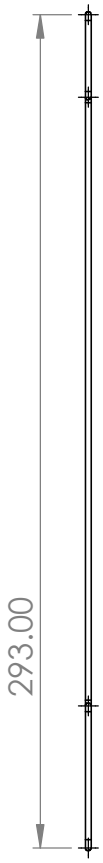


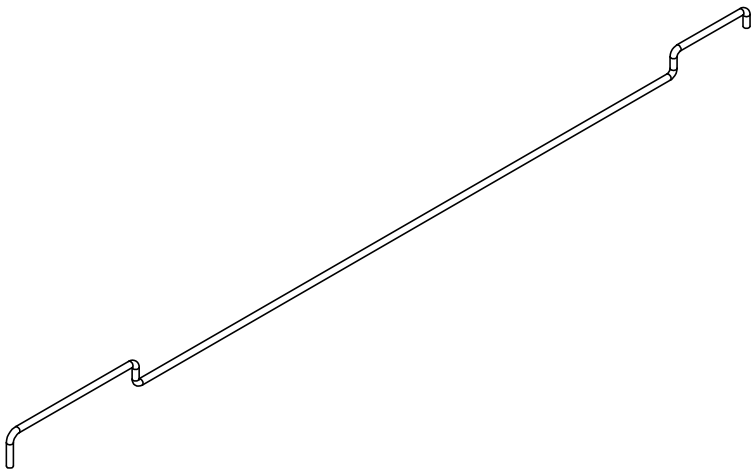
2. Feather position link

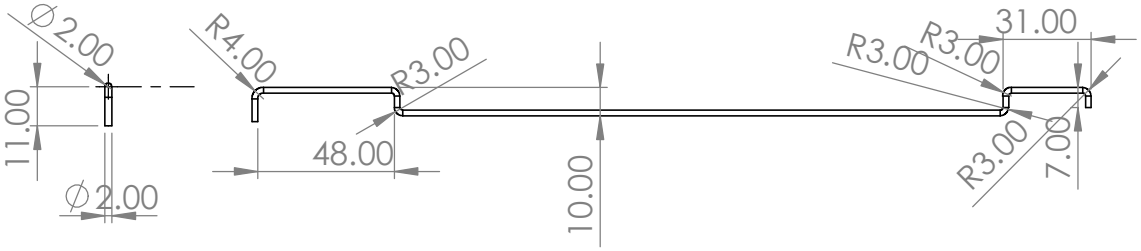


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE			TITLE:	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:		DWG NO.	A4
						Wing Assembly Parts	
				WEIGHT:		SCALE:1:5	SHEET 1 OF 1

3. Connecting Rod

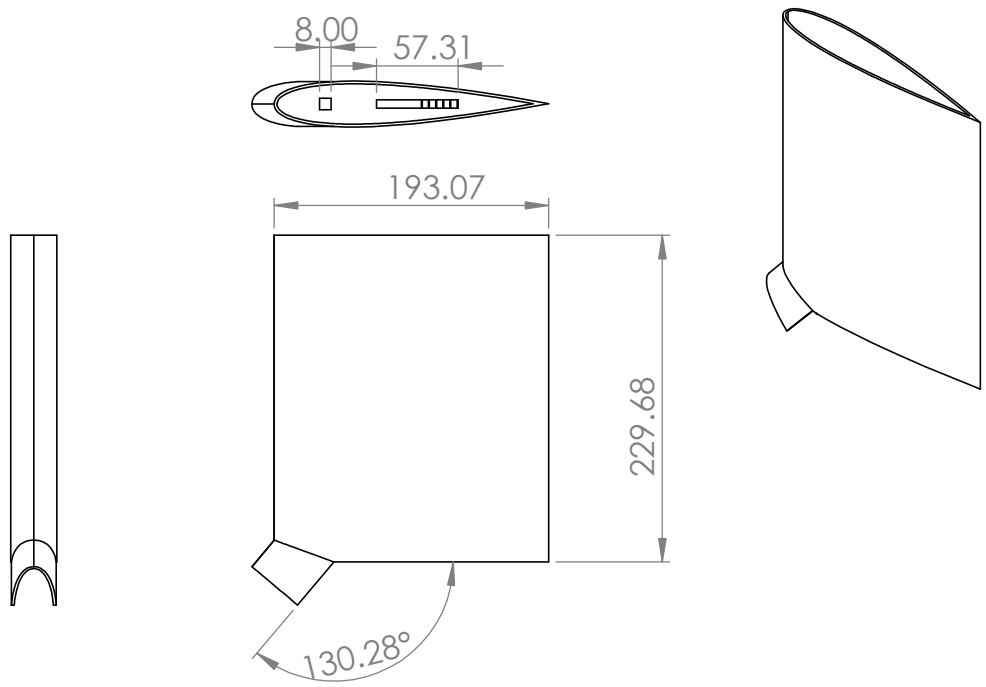




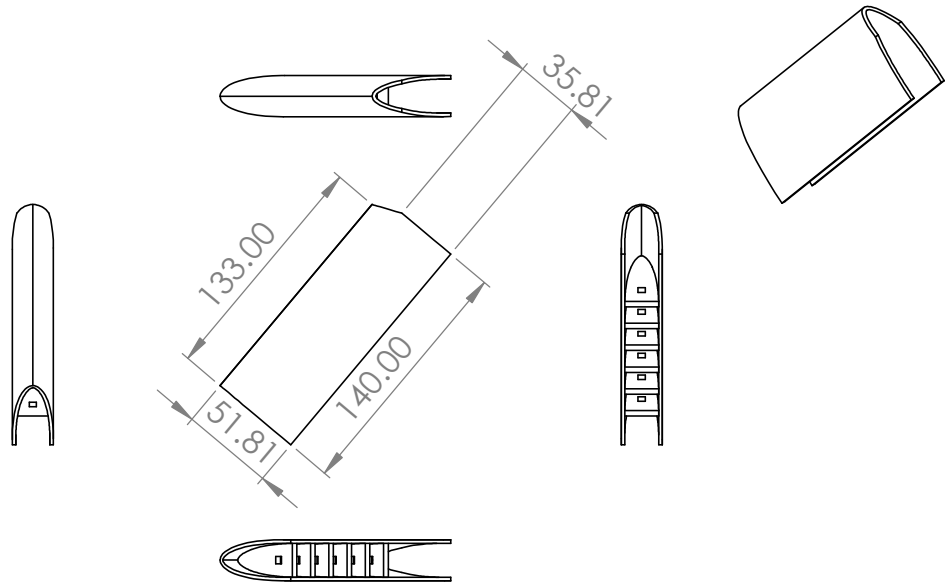


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION	
	NAME	SIGNATURE	DATE	TITLE:		
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL:	DWG NO. Connectiong rod	A4
				WEIGHT:	SCALE:1:5	SHEET 1 OF 1

4. AeroFoil (Naca 2412)

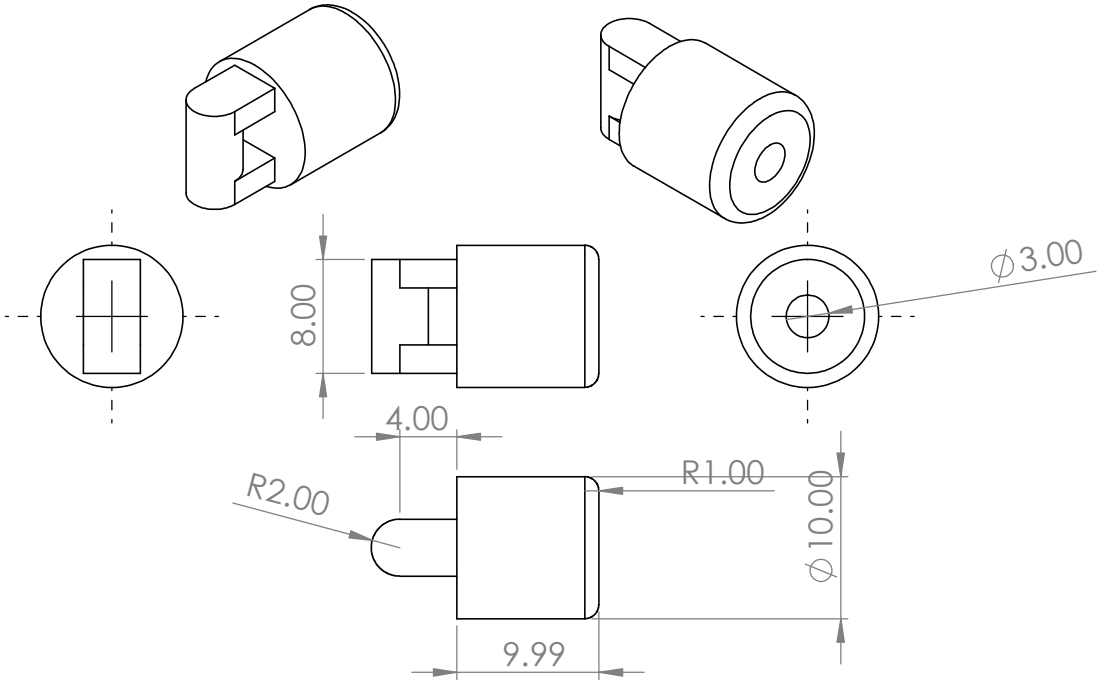


5. Feather Link Aerofoil (NACA 2412)

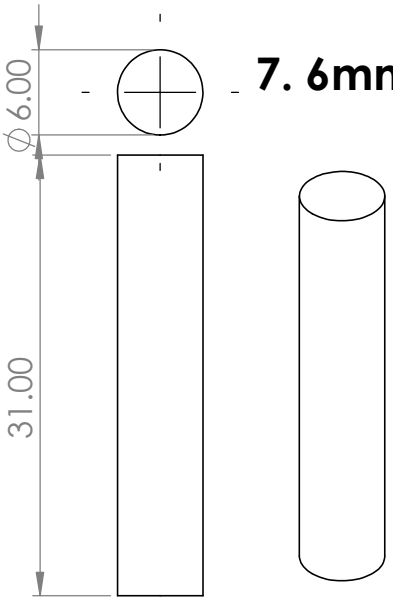


UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:		DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING	REVISION		
	NAME	SIGNATURE	DATE				TITLE:
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL:			DWG NO. Aerofoil
							A4
				WEIGHT:		SCALE:1:5	SHEET 1 OF 1

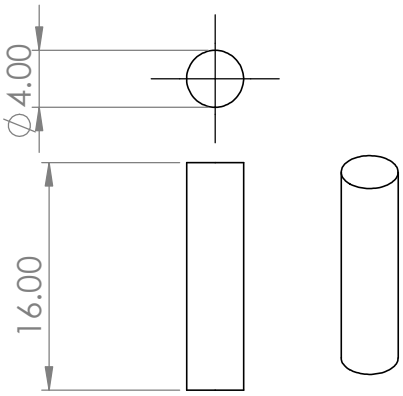
6. Wings - Feather Holder



7. 6mm Shaft



8. 4mm Shaft



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:	FINISH:	DEBURR AND BREAK SHARP EDGES	DO NOT SCALE DRAWING / REVISION

	NAME	SIGNATURE	DATE
DRAWN			
CHK'D			
APPV'D			
MFG			
Q.A			

TITLE: Wing Assembly Parts 2

MATERIAL:

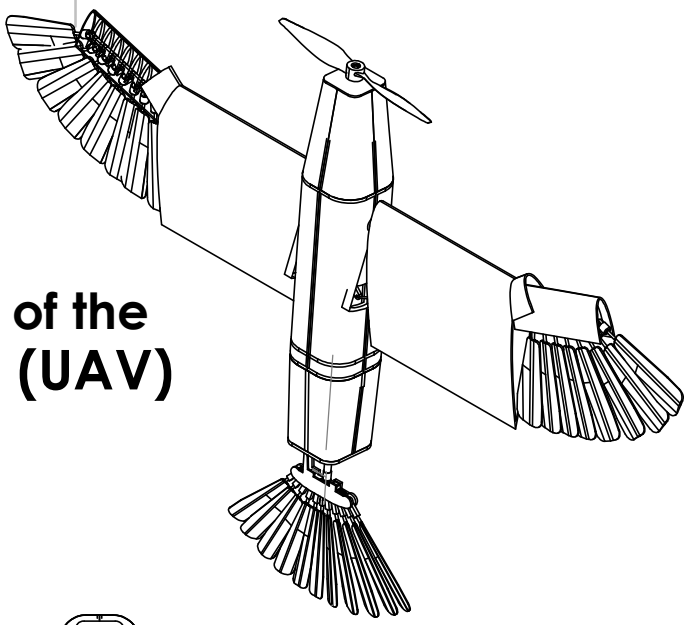
DWG NO. A4

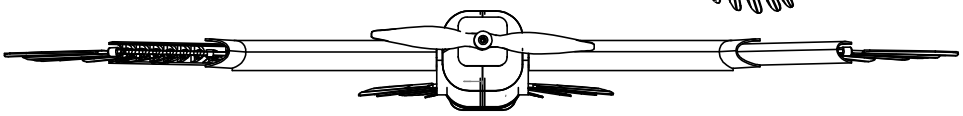
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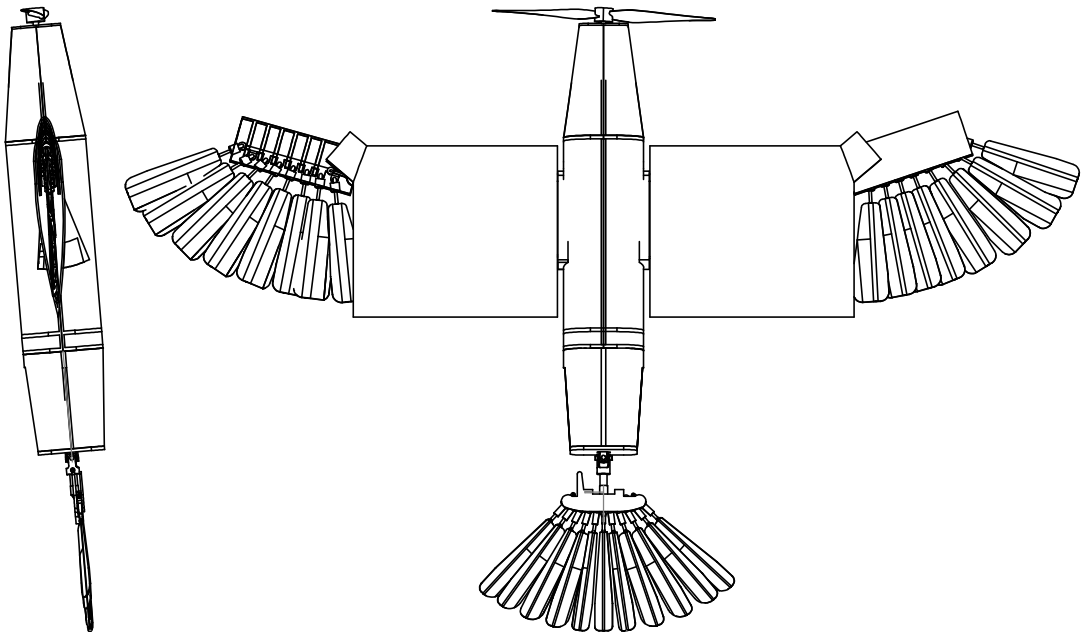
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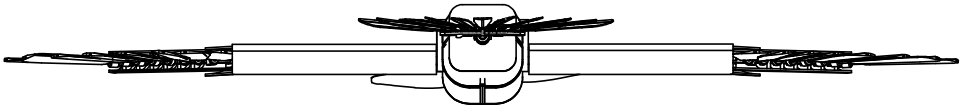
SHEET 1 OF 1

Complete Assembly of the Bird Wing Morphed (UAV)









UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: LINEAR: ANGULAR:		FINISH:			DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION
	NAME	SIGNATURE	DATE				TITLE:	
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Q.A				MATERIAL:			DWG NO.	A4
							Complete Assembly	
				WEIGHT:			SCALE:1:50	SHEET 1 OF 1

8. WING LAYOUT

The wing layout of the morphing wing is designed by studying the structure and movement of bird wings. The wing is based on the NACA 2412 air foil profile, which provides good lift characteristics and aerodynamic efficiency. The wing structure is divided into three main sections: the inner fixed section, the middle twisting section, and the outer span-extension section.

The inner fixed section is connected directly to the fuselage and provides structural support to the entire wing system. This section remains rigid during flight and helps in maintaining stability and load distribution. The middle section of the wing is designed for twist morphing, where the wing can twist up to 10° depending on flight conditions. Twist morphing improves roll control and aerodynamic performance during manoeuvring.

The outer wing section is designed for span morphing. A sliding spar mechanism allows the wingtip to extend up to 20% of the original wingspan. Increasing the span helps reduce induced drag during low-speed flight and improves lift generation. During high-speed flight, the wing can retract to reduce drag and improve stability.

The wing has a maximum thickness of 20 mm and an approximate chord length of 129.19 mm. The overall wing layout is designed to improve aerodynamic efficiency, flight stability, manoeuvrability, and adaptability under different flight conditions. The design combines bio-inspired concepts with modern aerospace engineering techniques to achieve efficient morphing wing performance.

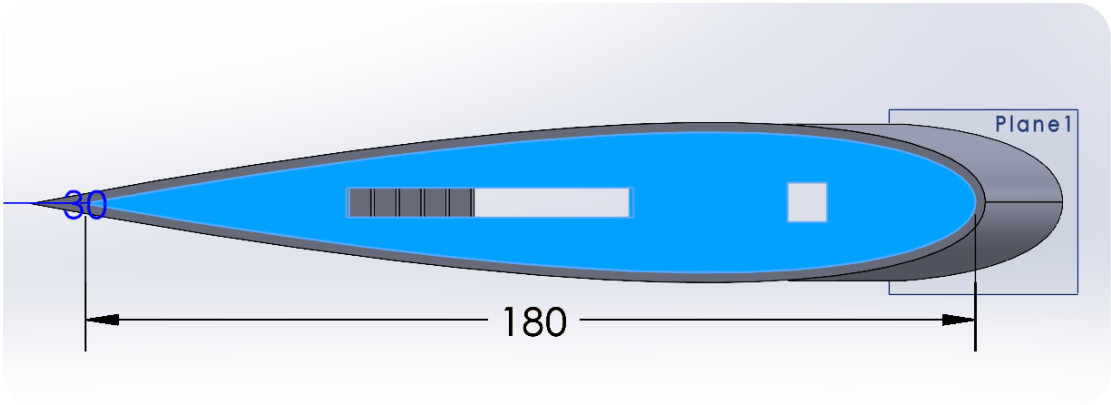


Fig. 8.1: NACA 2412 airfoil profile

9. BIRD WING MORPHING UAV FLIGHT PRINCIPLE

A Bird-Wing Morphing UAV is an unmanned aerial vehicle that changes the shape of its wings during flight, like how birds adjust their wings and tail. Instead of relying only on fixed wings and conventional control surfaces (ailerons, elevators, rudders), a morphing UAV modifies its wing geometry to control lift, drag, stability, and maneuverability.

9.1 Basic Flight Principle

Birds continuously adapt their wings to different flight conditions:

- **Extend wings** for maximum lift during takeoff and slow flight.
- **Retract or sweep wings** to reduce drag during fast flight.
- **Twist wings** to control roll and improve aerodynamic efficiency.
- **Change camber (curvature)** to optimize lift-to-drag ratio.
- **Adjust tail shape** for stability and maneuvering.

Morphing UAVs imitate these behaviors using actuators, flexible structures, and lightweight materials.

9.2 Main Morphing Mechanisms

1. Wingspan Morphing

The UAV increases or decreases wingspan.

- **Long span:** Higher lift, lower induced drag, better endurance.
- **Short span:** Higher agility, faster turns, reduced structural loads.

Like how soaring birds spread their wings while gliding.

2. Wing Sweep Morphing

The wing moves forward or backward relative to the fuselage.

- **Forward sweep:** Better low-speed lift.
- **Backward sweep:** Reduced drag at higher speeds.

This allows one aircraft to perform efficiently across a wider speed range.

3. Wing Twist Morphing

Different parts of the wing rotate slightly.

- One wing can generate more lift than the other.
- Produces roll motion without large aileron deflections.
- Improves aerodynamic efficiency.

This is like the wing-warping concept used in early aircraft.

4. Camber Morphing

The wing curvature changes smoothly.

- Increased camber → more lift for takeoff and climbing.
- Reduced camber → less drag for cruising.

Unlike flaps, camber morphing maintains a smoother aerodynamic surface.

5. Feather-Inspired Morphing

Artificial "feathers" fold or spread.

- Reduces wing area during high-speed flight.
- Increases maneuverability.
- Can assist roll control by asymmetric folding.

Researchers have demonstrated drones that fold outer wing sections similarly to bird primary feathers.

9.3 Flight Control

Instead of only using conventional control surfaces:

Motion	Morphing Method
Roll	Asymmetric wing folding or twisting
Pitch	Wing sweep/camber changes and tail morphing
Yaw	Differential wing geometry and tail adjustments
Speed Control	Wing area and sweep modification

10. MORPHING MECHANISM

The morphing mechanism is one of the most important parts of the bird-inspired wing system. It is designed to imitate the movement and flexibility of bird wings and tail feathers during flight. Birds continuously adjust the position, span, twist, and orientation of their wings to achieve stable and efficient flight under different aerodynamic conditions. Based on this concept, a mechanical morphing system is developed to reproduce similar motions in an unmanned aerial vehicle (UAV).

The mechanism mainly consists of a central rotating shaft, linkage rods, servo motors, flexible ribs, sliding members, and a tail twisting arrangement. These components work together to provide controlled movement of both the wing and tail sections. The entire mechanism is modelled and assembled using SolidWorks CAD software to visualize the motion and interaction between components.

The main shaft acts as the backbone of the mechanism and supports the entire wing structure. It transfers motion from the actuator to the wing sections through mechanical linkages. Several pivot joints are connected along the shaft, allowing smooth angular movement similar to the spreading motion of bird feathers. These joints reduce mechanical stress and allow flexible movement during operation.

Servo motors are used as actuators to control different morphing actions. One servo motor control the wing spreading and folding motion, while another servo controls lateral motion with low power consumption. By changing the servo angle, the wing structure can expand or retract depending on flight requirements.



Fig. 10.1: Four Bar Linkage Morphing Mechanism

The wing morphing arrangement uses multiple connected ribs that act like bird feathers. These ribs are attached through small hinge joints, allowing gradual and smooth deformation of the wing surface. Flexible rods and tendon-like members are used to synchronize the movement of different sections of the wing. This arrangement reduces sudden deformation and maintains smooth aerodynamic flow over the wing surface.

A sliding spar mechanism is also incorporated into the design to achieve span morphing. The spar can slide linearly inside the wing structure, allowing the wingtip to extend outward during low-speed flight conditions. Increasing the wingspan improves lift generation and reduces induced drag. During high-speed flight, the wing retracts to reduce aerodynamic resistance and improve stability.

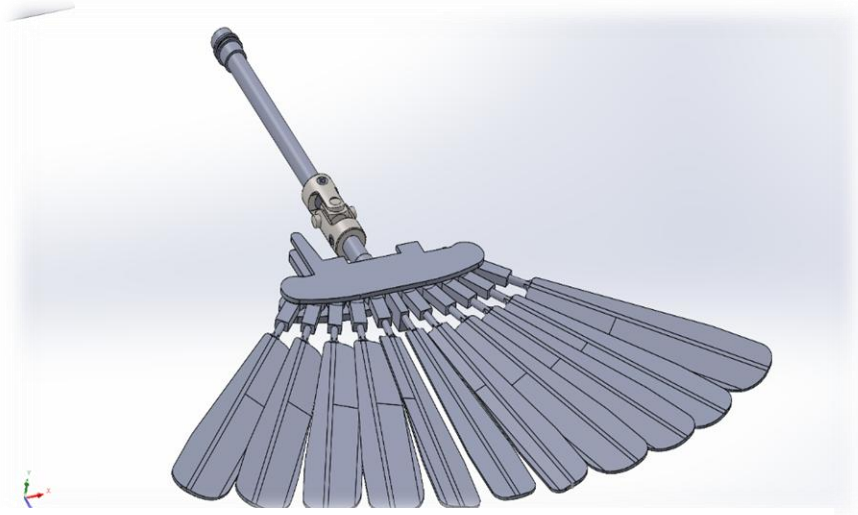


Fig. 10.2: Tail Twisting Mechanism

The tail morphing mechanism is designed to imitate the twisting and directional movement of bird tails. A universal joint arrangement is used near the tail section to allow multi-directional rotation. The tail twisting mechanism helps in pitch and yaw control of the UAV. This improves manoeuvrability, directional stability, and overall flight control during turning and gliding operations.

The complete morphing mechanism is designed to provide smooth feather-like motion while minimizing mechanical complexity. The bio-inspired mechanism improves lift generation, reduces drag, enhances manoeuvrability, and increases aerodynamic efficiency. Such morphing systems can play an important role in future UAV and aerospace applications where adaptive wing structures are required for efficient and stable flight.

11. CAD Model and Machine Design

The CAD model and machine design of the bird-inspired morphing RC aircraft are developed by combining aerodynamic calculations, biomimetic morphing concepts, lightweight structural materials, and advanced mechanical linkage systems. The design is inspired by the EPFL morphing drone research, where bird-like wing sweep, tail feather deployment, and multi-degree-of-freedom flight mechanisms are integrated into a compact unmanned aerial vehicle (UAV) platform.

The complete aircraft is designed using SolidWorks CAD software. The assembly includes the fuselage structure, four-bar linkage wing mechanism, central actuation shaft, morphing tail assembly, aerodynamic feather structures, lofted wing skin, servo motor mounts, internal structural supports, and flexible transition membranes. The CAD model helps visualize the movement, deformation, and interaction of all moving components before prototype fabrication.

11.1 Aircraft Geometry and Design Specifications

The aircraft is designed as a mini-class sport RC aircraft with a bird-inspired morphing mechanism. The geometry is optimized for manoeuvrability, aerodynamic efficiency, and lightweight construction.

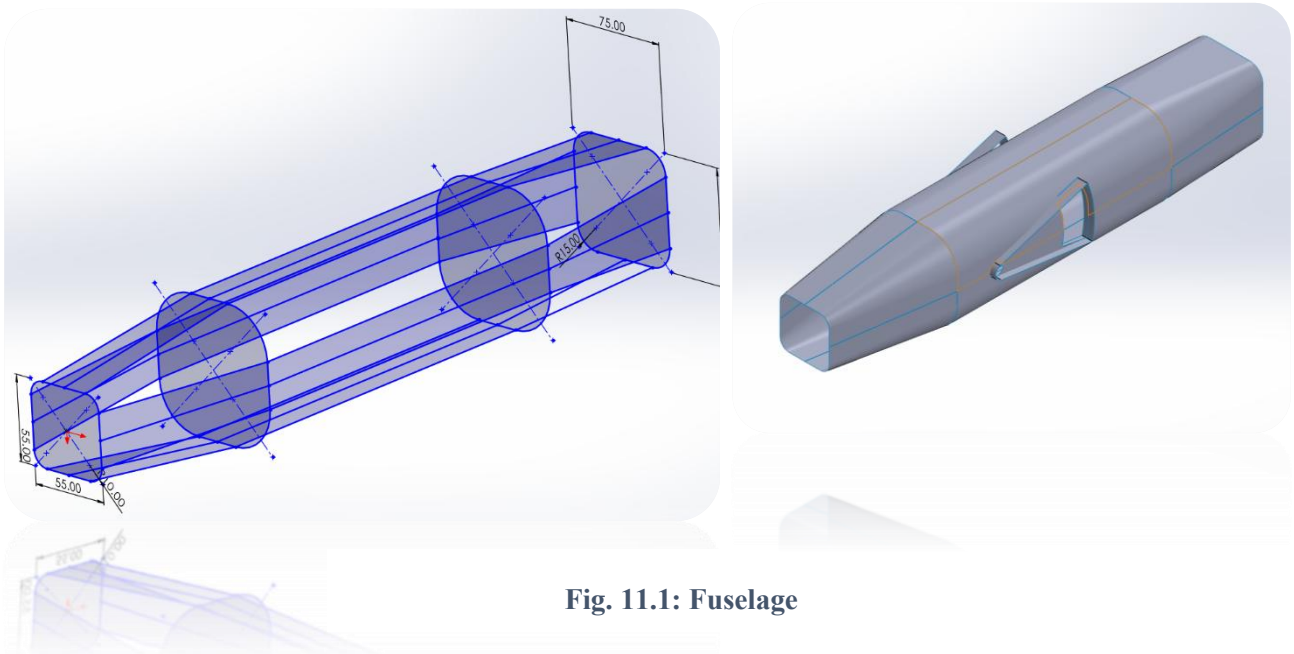


Fig. 11.1: Fuselage

Table 11.1: Main Aircraft Dimensions

Parameter	Dimension
Wingspan	800 mm
Fuselage Length	476 mm
Wing Chord	170 mm
Wing Thickness	27 mm
Wing Area	136,000 mm²
Tail Boom Length	165 mm

The aircraft uses a constant rectangular wing planform with a NACA 2412 airfoil profile. The selected airfoil provides stable aerodynamic characteristics and sufficient internal thickness for structural spars and servo integration.

11.2 Airfoil and Aerodynamic Design

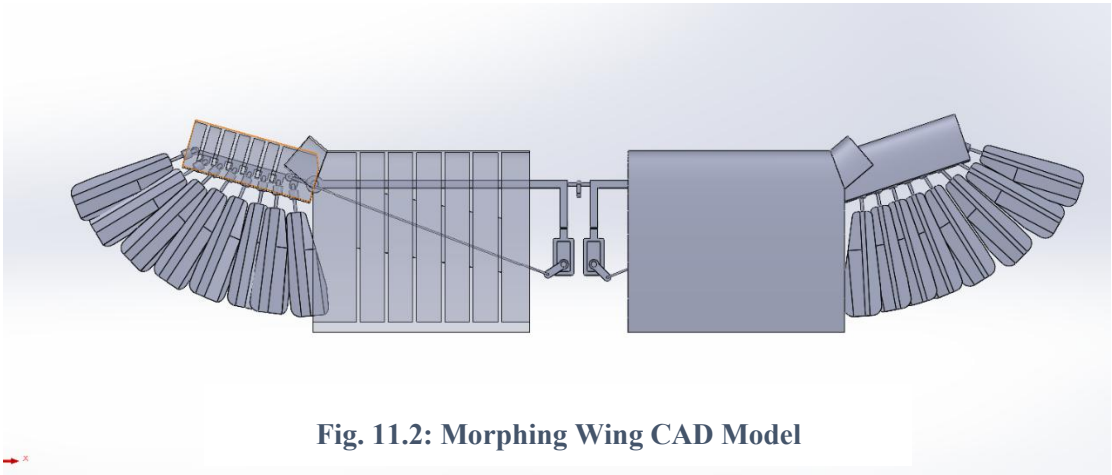


Fig. 11.2: Morphing Wing CAD Model

The wing profile is developed using a symmetric or semi-symmetric aerofoil with a thickness-to-chord ratio of approximately 13.1%. The airfoil thickness is maintained at 5.97 mm for the scaled design.

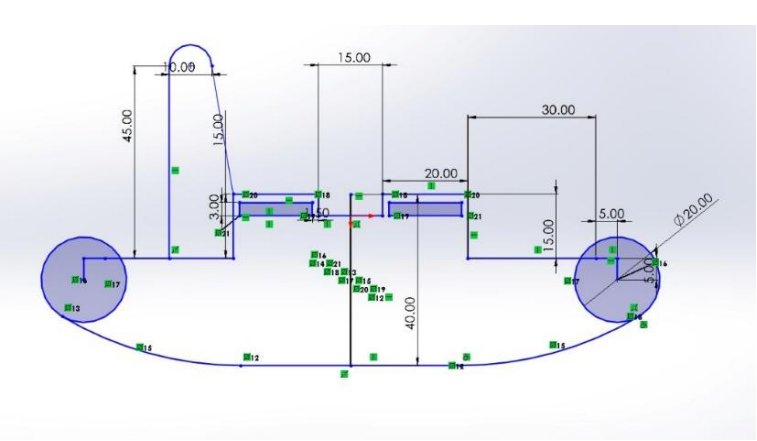
The selected airfoil improves lift generation at low speeds and provides stable aerodynamic during manoeuvres. Symmetric airfoils are particularly suitable for aerobatic operations because they provide similar aerodynamic performance during upright and inverted flight conditions.

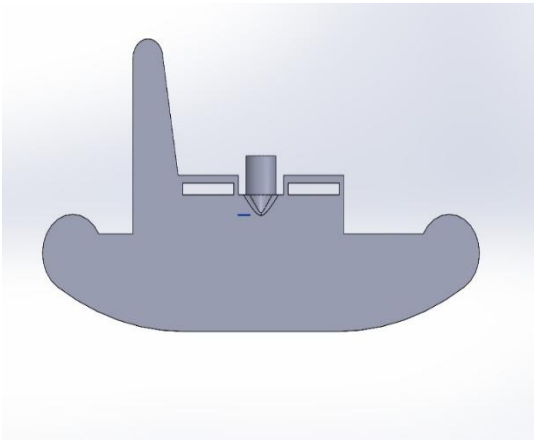
The aircraft has an aspect ratio of approximately 5.46, corresponding to a low-to-medium aspect ratio sport aircraft:

$$AR = \frac{b^2}{S}$$

This configuration improves roll response, structural rigidity, and manoeuvrability during dynamic flight operations.

11.3 Central Actuation Shaft and Tail Feather Mechanism





The central actuation shaft is one of the primary mechanical systems of the morphing aircraft. The shaft is positioned longitudinally inside the fuselage and supported using lightweight bulkheads and concentric support discs.

At the rear section of the shaft, a multi-hinge feather array is attached to imitate bird tail feathers. The hinge arrangement allows the tail feathers to spread symmetrically during operation. This fan-like movement improves yaw stability, directional control, and aerodynamic efficiency. The shaft transfers rotational motion from the servo motor to the tail feather mechanism, providing pitch control, yaw control, and adaptive tail incidence like natural bird flight.

11.4 Four-Bar Wing Sweep Linkage Mechanism

The wing morphing system uses a four-bar linkage mechanism composed of push rods, pivot joints, and an actuation crank. This mechanism converts rotational servo motion into controlled wing sweeping movement.

The linkage mechanism allows the wing structure to sweep dynamically between approximately 45° and 130°.

Low-speed flight: The wings extend outward to maximize lift generation.

High-speed operation: The wings retract backward to reduce drag and improve aerodynamic stability.

Revolute joints are used at linkage pivots to reduce friction and maintain smooth motion. The four-bar linkage provides synchronized movement of both wing sections while minimizing mechanical complexity.

11.5 Aerodynamic Feather Profiles

The wing and tail structures consist of multiple aerodynamic feather profiles designed using loft and surface modelling techniques. Each feather contains a tapered aerodynamic geometry with rounded fillets and cylindrical root sleeves.

The cylindrical root sleeve allows secure attachment of each feather to structural spars or linkage pins. Slight variations in feather geometry create aerodynamic twist and allow overlapping during wing retraction. These feather structures imitate the motion of natural bird feathers and improve airflow management during morphing operations. Smooth overlapping of the feathers reduces airflow separation and aerodynamic drag.

11.6 Surface-Modelled Fuselage Structure

The fuselage structure is developed using SolidWorks surface modelling techniques rather than conventional solid extrusion. Boundary surfaces are used to create a lightweight hollow shell with smooth aerodynamic contours.

The fuselage houses internal components such as:

Batteries and Electronic Speed Controllers (ESCs)

Servo motors and linkage systems

Central shafts and internal wiring assemblies

Surface modelling helps reduce structural weight while maximizing internal component space, which also improves manufacturability for 3D printing and composite fabrication methods.

11.7 Lofted Wing Skin and Flexible Transition Membrane

A lofted wing skin is developed between the rigid inner wing structure and the outer morphing feather assemblies. This transition membrane acts as a flexible aerodynamic surface that maintains smooth airflow during wing sweeping and twisting operations.

The flexible skin undergoes controlled deformation during morphing movements without producing major surface discontinuities. Maintaining aerodynamic continuity is important for reducing drag and improving flight efficiency.

11.8 Structural Material Selection

The aircraft structure uses lightweight composite materials to reduce total mass while maintaining structural rigidity. Carbon Fibre Reinforced Polymer (CFRP) rods are selected for the primary wing spars and internal support members.

Table 11.2: Material Allocation Matrix

Component	Material	Mechanical Function
Main Wing Spar	Carbon Fibre	Primary load-carrying beam
Wing Ribs	PLA / PETG	Lightweight structural support
Feather Structures	PLA / Nylon	Morphing aerodynamic surface
Fuselage Shell	Composite Surface Structure	Lightweight aerodynamic body

Component	Material	Mechanical Function
Ailerons	Thin PLA / Balsa	Lightweight control surfaces
Motor Firewall	Aluminium Alloy	Vibration and thrust resistance

Carbon fibre spars resist bending loads generated during high-G manoeuvres. PLA and PETG materials are used for 3D printed components because of their lightweight properties and ease of fabrication.

11.9 Control Surface and Tail Design

Dual ailerons are positioned along the trailing edge of the wing to improve roll control and manoeuvrability. The horizontal stabilizer and elevator provide pitch stability, while the vertical stabilizer improves yaw control and directional stability.

Table 11.3: Control Surface Dimensions

Component	Dimension
Aileron Length	-
Aileron Width	-
Horizontal Stabilizer Area	5,742 mm²
Elevator Area	1,451.5 mm²
Vertical Stabilizer Area	2,290 mm²

11.10 Servo Motor and Internal Mechanical Integration

Micro servo motors are integrated inside the wing root and fuselage structure. The available wing thickness provides sufficient internal clearance for routing push rods, linkage assemblies, and wiring channels.

Servo motors actuate the four-bar linkage mechanism, tail feather deployment system, and control surfaces. Proper mounting brackets and structural support are incorporated to reduce vibration and maintain mechanical alignment.

11.11 CAD Assembly and Motion Simulation

The complete aircraft assembly is created using SolidWorks CAD software. Mechanical mates and motion constraints are applied between linkage joints, hinges, shafts, and servo connections.

Motion simulation is performed to study wing sweeping, feather deployment, tail twisting, and aerodynamic deformation during morphing operations. The CAD environment helps identify interference between moving components and improves overall mechanical synchronization.

12. EXPECTED OUTCOMES

This study is expected to provide a clear understanding of bird-inspired wing morphing concepts and their application in modern aircraft. By analysing different morphing techniques, the project aims to highlight improvements in aerodynamic performance, efficiency, and control.

12.1 Understanding of Morphing Concepts

The project will help in understanding how birds adjust their wings during flight and how these mechanisms can be applied to aircraft design. It includes the study of different morphing types such as sweep, camber, and twist, and their role in flight performance.

12.2 Aerodynamic Performance Improvement

By applying morphing techniques, the aircraft is expected to show improved aerodynamic performance. Changes in wing shape can increase lift, reduce drag, and improve the lift-to-drag (L/D) ratio, resulting in better efficiency across different flight conditions.

12.3 Stability and Control

Morphing wings are expected to enhance the stability and control of the aircraft. By adjusting the wing shape, the aircraft can respond better to different flight conditions and maintain balanced and smooth flight.

12.4 Material and Mechanism Selection

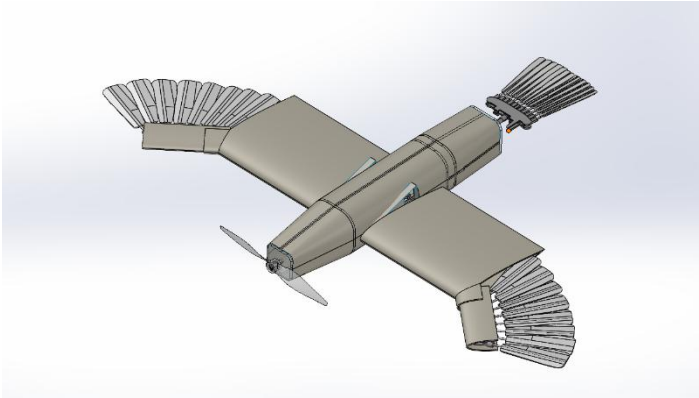
The study will identify suitable materials such as composite materials and smart materials like Shape Memory Alloys (SMA). It will also provide an understanding of different actuation mechanisms used to achieve morphing.

12.5 Conceptual Design Development

The project is expected to result in a basic conceptual design of a morphing wing system. This design will combine theoretical knowledge with practical considerations for future development.

13. ANALYSIS OF THE BIRD WING MORPHED UAV

13.1 Analysis of Normal flight conditions



Simulation Report Summary: Normal Flight Conditions

This report details a steady-state fluid flow simulation conducted on the assembly [t1.SLDASM] under Normal flight conditions using an external computational domain (1.566 m × 0.630 m x 1.425 m). The analysis evaluates a laminar and turbulent air flow at an ambient temperature of 303.15 K (30°C) and standard static pressure (101, 325 Pa), with a default ambient velocity vector of 7.0 m/s moving purely along the Z-axis. Gravitational effects are enabled along the negative Y-axis at −9.81 m/s2.

The converged solver metrics extracted from the Goal Plot 1.xlsx spreadsheet provide key insight into the model's aerodynamic performance:

Velocity Profiles: The global average velocity stabilizes at 6.92 m/s, peaking at a maximum velocity of 8.04 m/s around the geometry. The dominant flow travels in the Z- direction (average GG Velocity (Z) = 6.92 m/s), while minor recirculation/lateral dispersion is captured by minimal and maximal velocities along the X (−2.23 to 1.97 m/s) and Y (-2.52 to 2.47 m/s) axes.

Pressure Characteristics: The static/dynamic balance yields an average total pressure of 101, 343.87 Pa, with a localized minimum

Aerodynamic Forces: The structural model experiences a net aerodynamic lift or vertical force (GG Force (Y)) of 0.0132 N and an opposing skin friction drag force (GG Friction Force (Z)) of 0.0337 N.

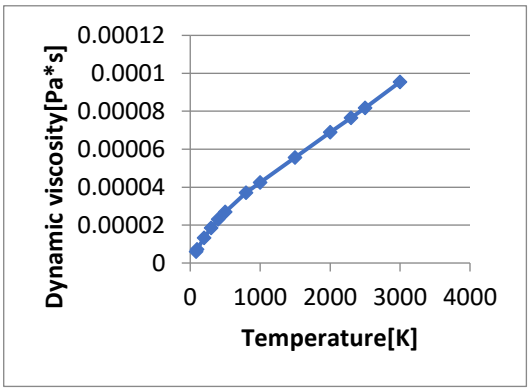
Engineering Database

Gases

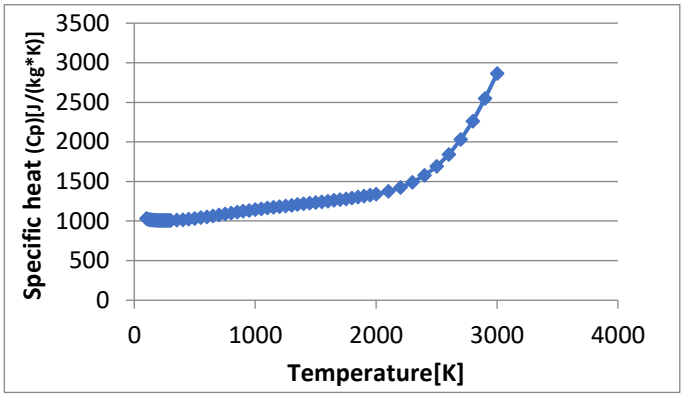
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Molecular mass: 0.0290 kg/mol

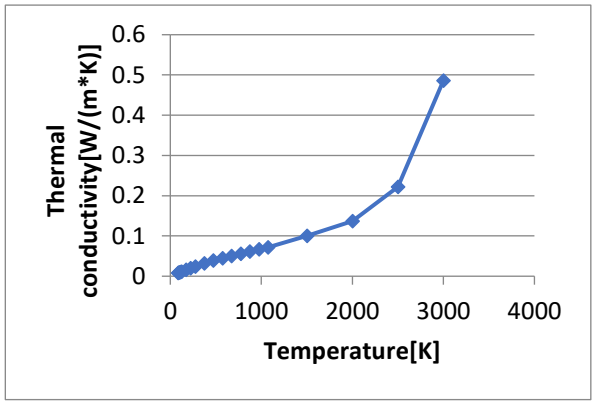
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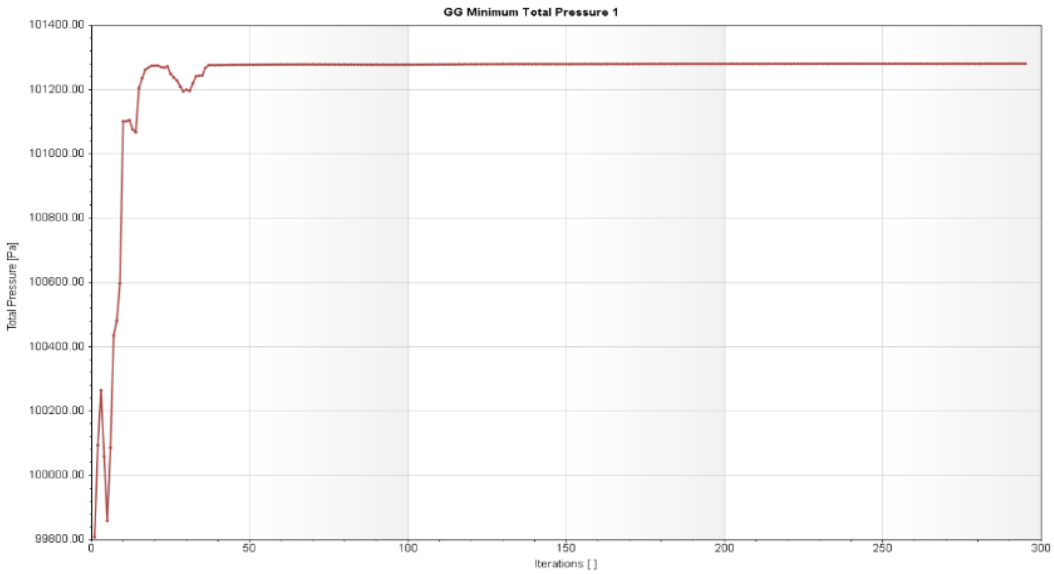
Specific heat (Cp):



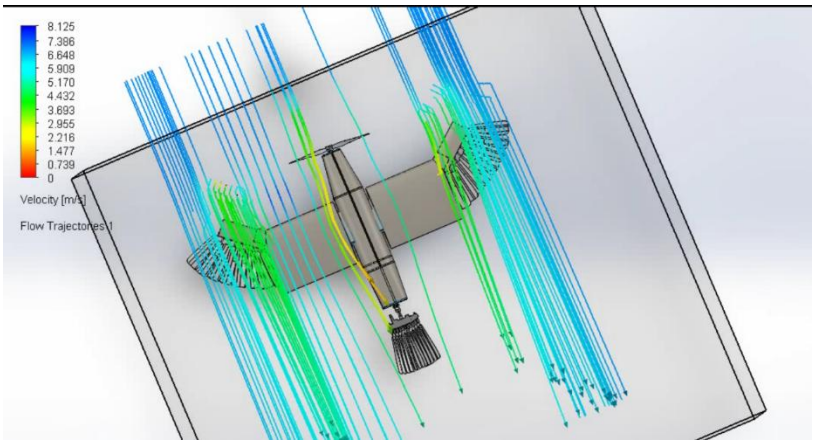
Thermal Conductivity



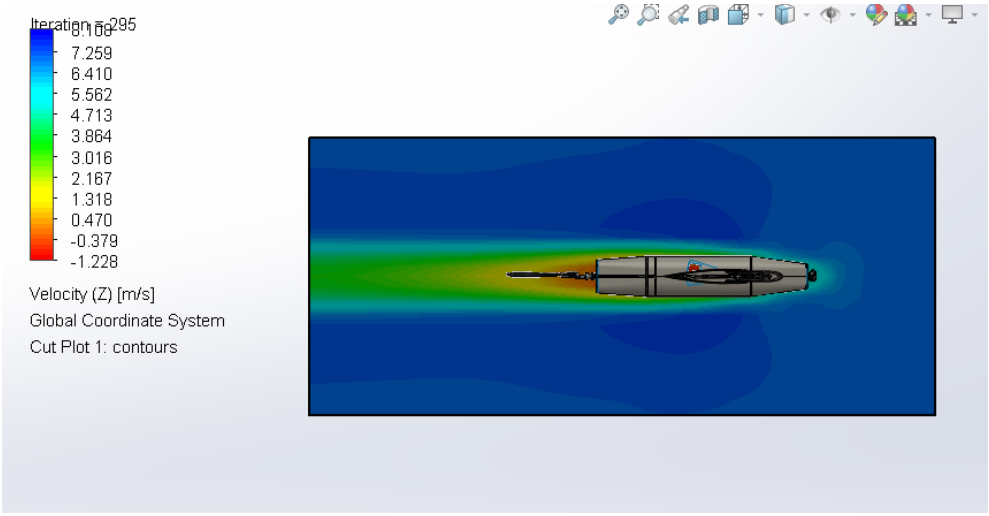
Minimum Total Pressure



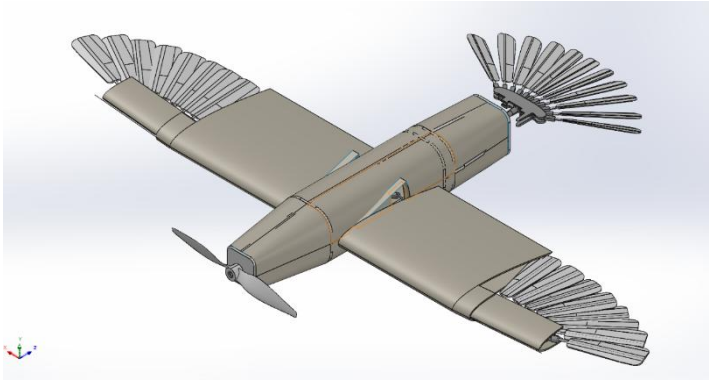
Fluid {Air} Acting on the model {Normal Condition}



Velocity of air Acting on the Model {Normal Condition}



13.2 Analysis of Normal flight conditions with Tail Part Open.



This report documents the setup and parameters for a steady-state Computational Fluid Dynamics (CFD) simulation conducted within a defined 3D computational domain measuring 1.627 m × 0.630 m × 1.423 m along the X, Y, and Z axes, respectively. The analysis evaluates a laminar and turbulent flow regime using Air as the fluid medium, under standard thermodynamic ambient conditions consisting of a static

pressure of 101, 325.00 Pa and a temperature of 303.15 K (30°C). Gravitational effects are actively factored into the model along the negative Y-axis (−9.81 m/s2), while default wall conditions are designated as adiabatic with zero roughness. The external flow field is driven by a uniform incoming velocity vector of 12.000 m/s moving strictly in the positive Z direction, accompanied by a low turbulence intensity of 0.10%.

To monitor the simulation's progress and ensure numerical accuracy, 17 distinct global goals have been established to track the minimum, average, and maximum thresholds for both total pressure and velocity components across all three coordinate axes. Additionally, vital physical metrics such as the lift/downforce component via Force (Y) and the aerodynamic resistance via Friction Force (Z) are continuously evaluated for convergence. The tracking, convergence trajectories, and final data points for these critical performance metrics are comprehensively recorded in the accompanying file "Goal Plot 1.xlsx". The solver is configured to terminate operations upon satisfying convergence criteria across an analysis interval of 0.5, or upon reaching a maximum limit of 4 travels, ensuring an optimized and stable engineering solution.

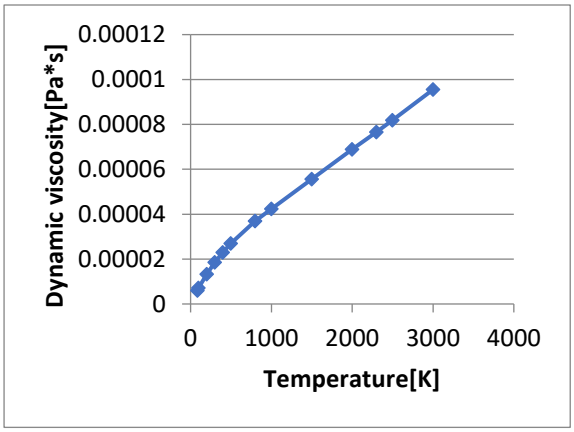
Engineering Database

Gases

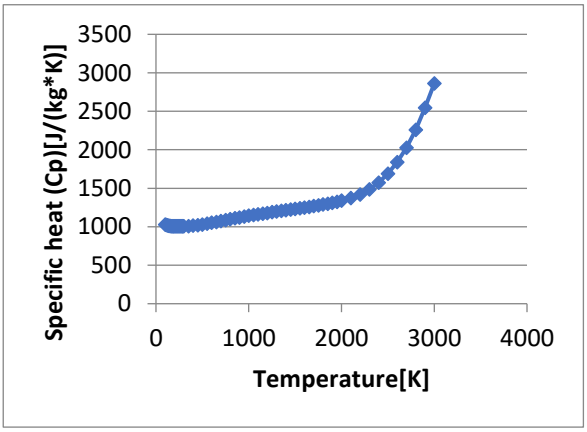
Air

Path: Gases pre-defined
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Molecular mass: 0.0290 kg/mol

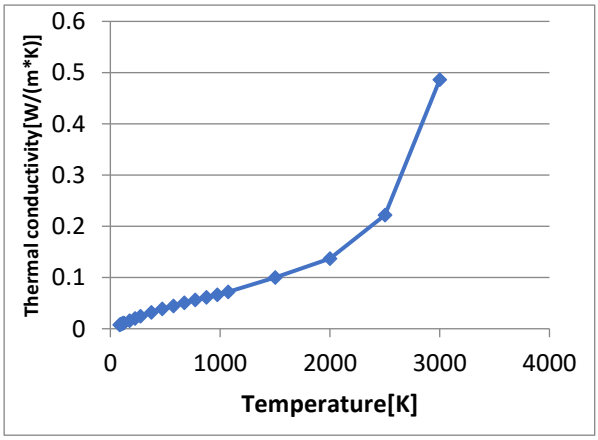
Dynamic viscosity



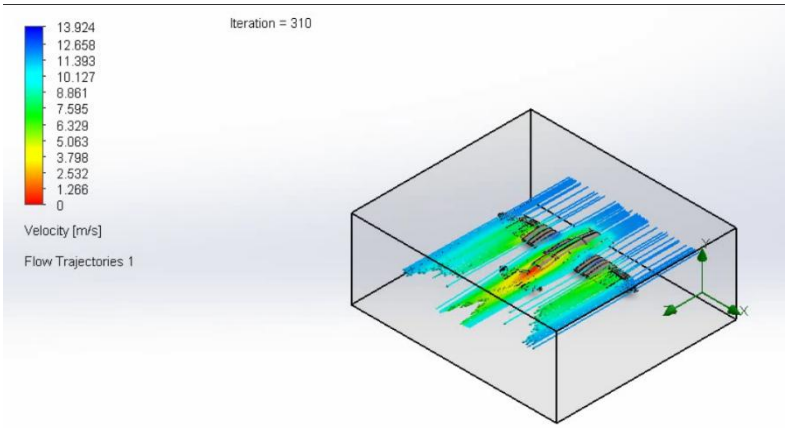
Specific heat (Cp)



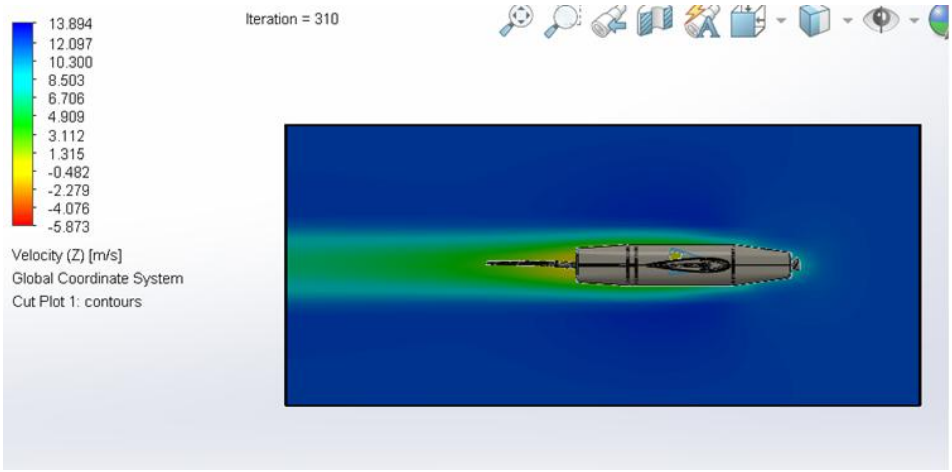
Thermal conductivity



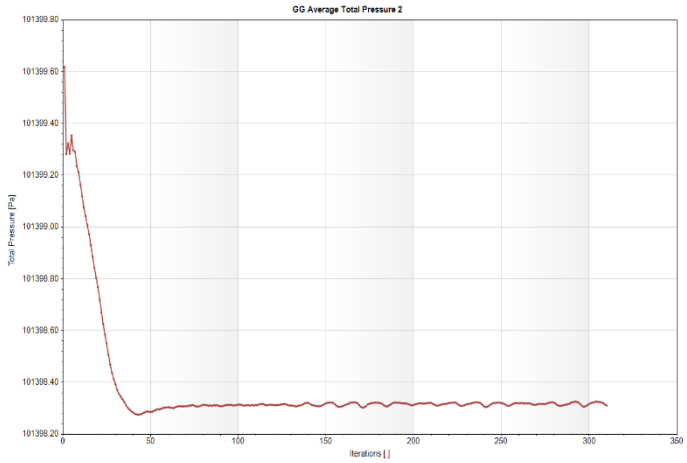
Fluid {Air} Acting on the model {Normal Condition}



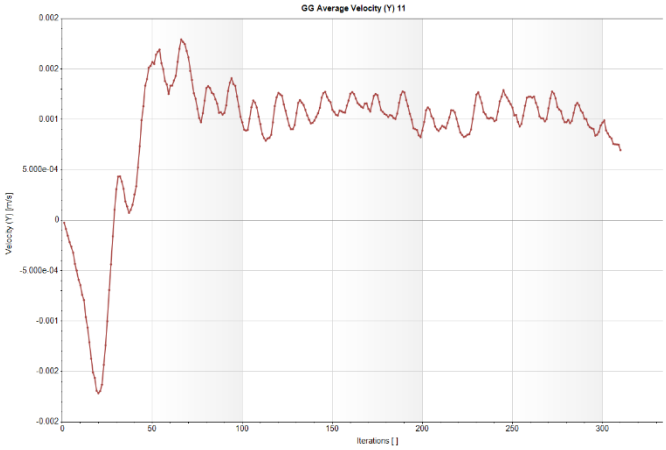
Velocity of air Acting on the Model {Normal Condition with Tail Part Open}



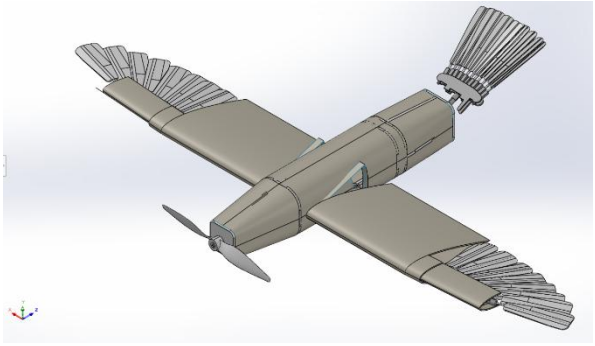
Average Total Pressure Graph



Average Velocity at Y (11).



13.3 Analysis of LIFT Condition.



The CFD simulation for the aircraft design, detailed in the file "report.docx" and quantified in "Goal Plot 1.xlsx", demonstrates a stable and converged aerodynamic solution. The simulation successfully reached 100% convergence progress, indicating that the computational model is numerically robust and consistent. Key performance metrics extracted from "Goal Plot 1.xlsx" reveal a lift force (Force Y) of approximately 3.15 N and a frictional drag force (Friction Force Z) of approximately 0.09 N. Given the low drag-to-lift ratio (0.09/3.15 ≈ 0.028), the aircraft displays promising aerodynamic efficiency, suggesting it is well-suited for stable flight within the simulated laminar and turbulent flow conditions. While these metrics point to a positive aerodynamic profile, determining if the aircraft is "better" to fly ultimately depends on the specific mission requirements and flight envelope targets established in the project design phase.

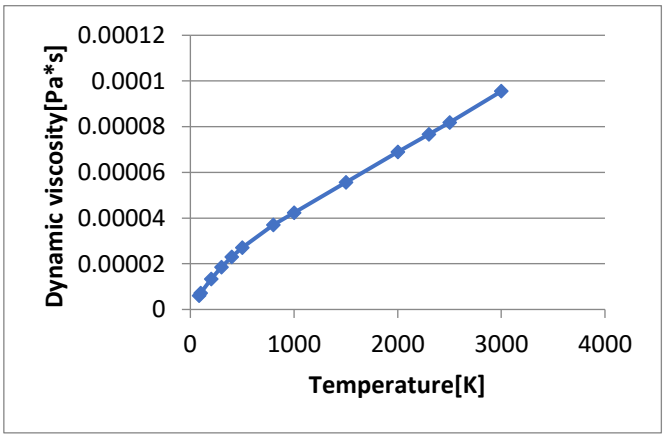
Engineering Database

Gases

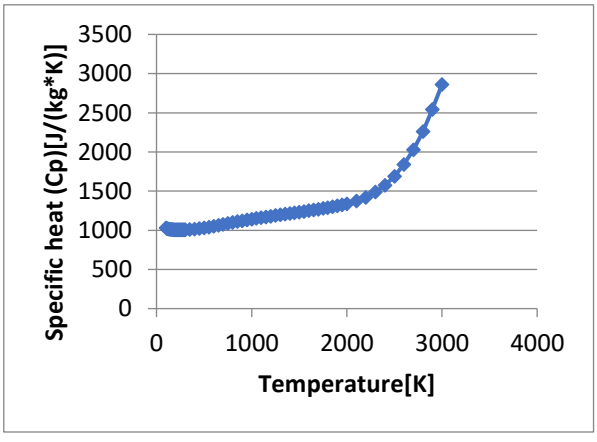
Air

Path: Gases pre-defined
Specific heat ratio (Cp/Cv): 1.399
Molecular mass: 0.0290 kg/mol

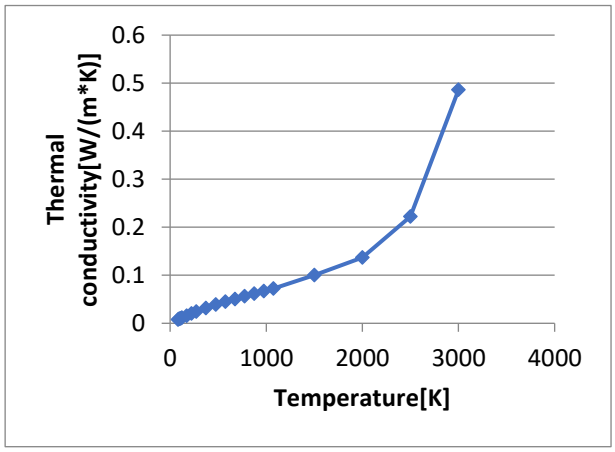
Dynamic viscosity



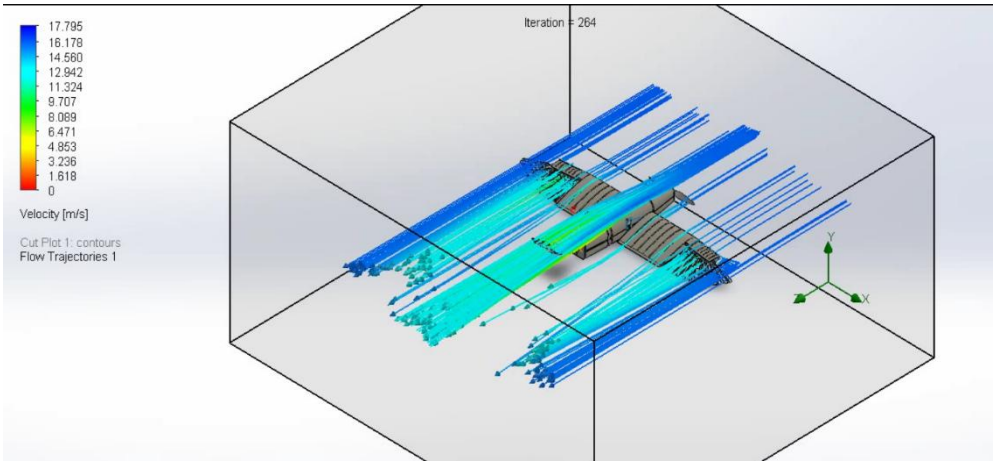
Specific heat (Cp)



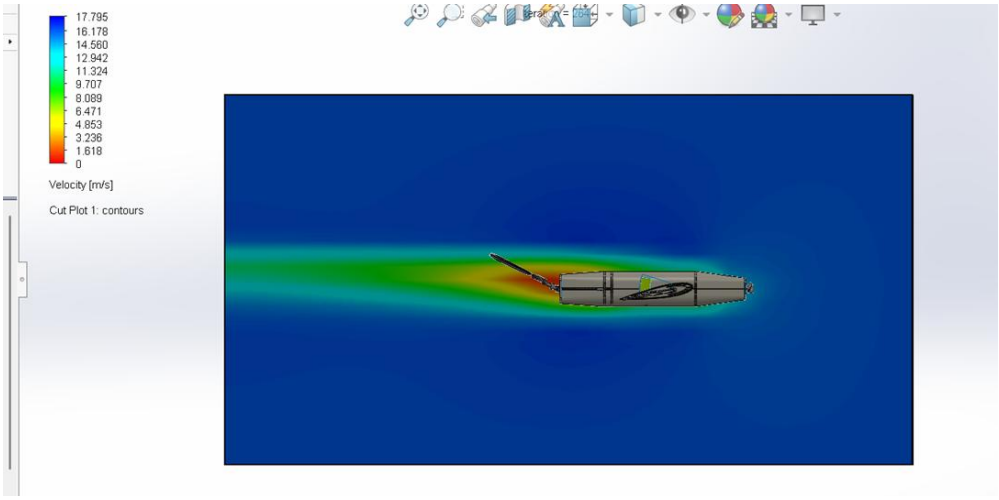
Thermal conductivity



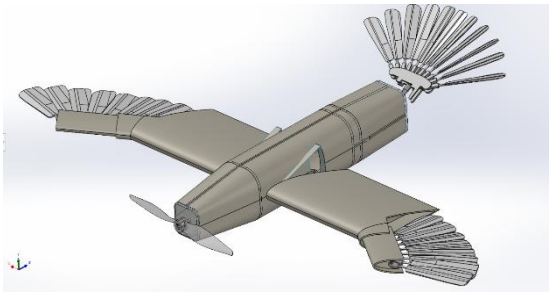
Fluid {Air} Acting on the model {For LIFT Condition}



Velocity of air Acting on the Model {For LIFT Condition}



13.4 Analysis of LIFT Condition with Tail Open.

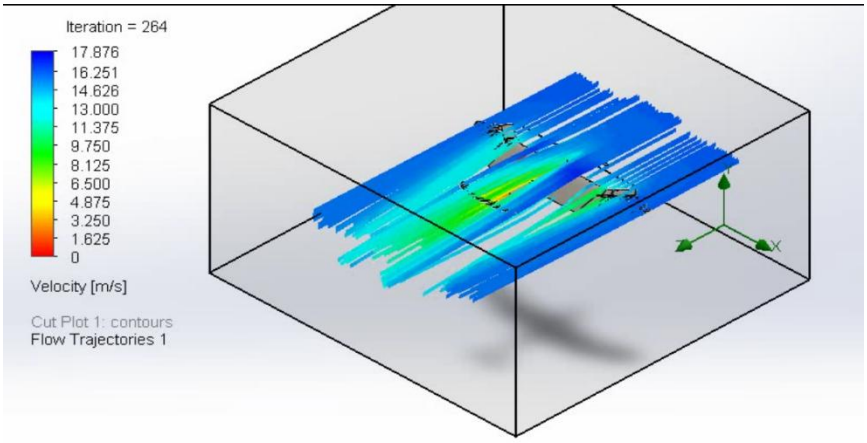


The simulation results indicate a convergent aerodynamic analysis operating at an average flow velocity of approximately 15.90 m/s primarily directed along the Z-axis (fluid flow direction). The simulation captures vital surface and fluid interactions, showing a maximum overall velocity of 18.13 m/s and minimal local backflows or recirculation (indicated by the small negative X and Y localized velocity components). Mechanically, the fluid interaction yields a net Normal/Lift Force (GG Force (Y) 16) of 2.846 N and a localized Friction Force (GG Friction Force (Z) 17) of 0.104 N.

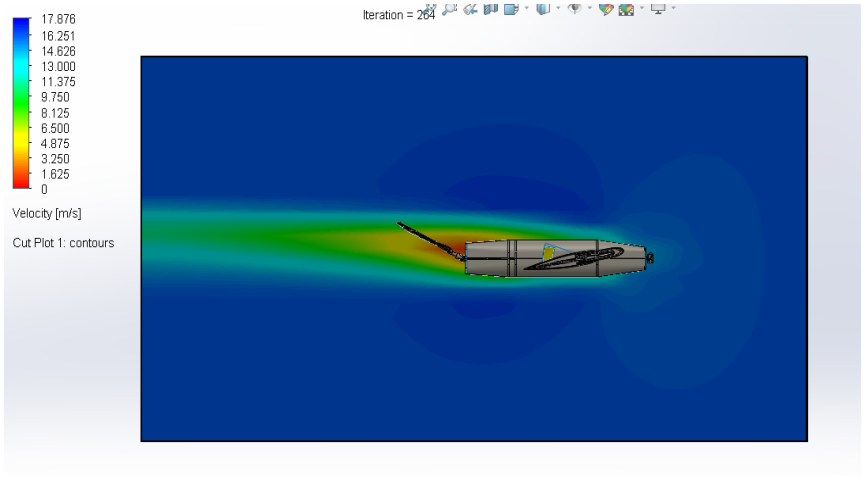
Whether this aircraft will fly depends strictly on its total physical mass:

Yes, it will fly if the total weight of this aircraft model (including its battery, servos, structure, and smart components) is less than 290 grams (equivalent to≈ 2.84 N of gravitational force).

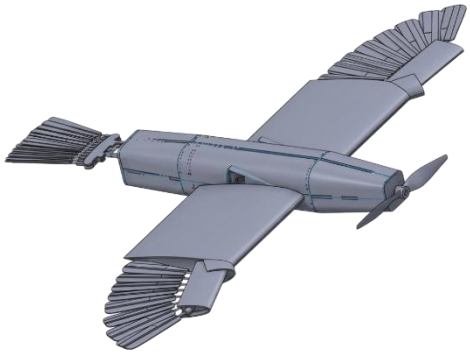
Fluid {Air} Acting on the model {For LIFT Condition with Tail Open}



Velocity of air Acting on the Model {For LIFT with Tail Open Condition}



13.5 Analysis of CRUISING Condition.



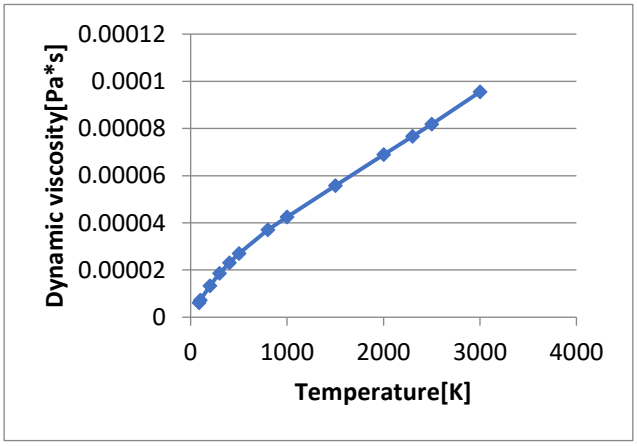
Engineering Database

Gases

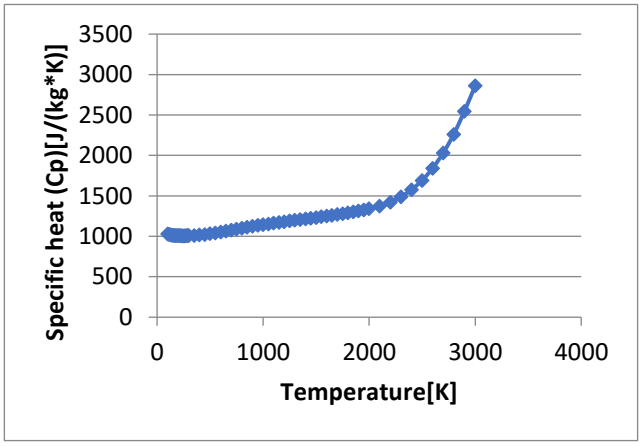
Air

Path: Gases pre-defined
Specific heat ratio (Cp/Cv): 1.399
Molecular mass: 0.0290 kg/mol

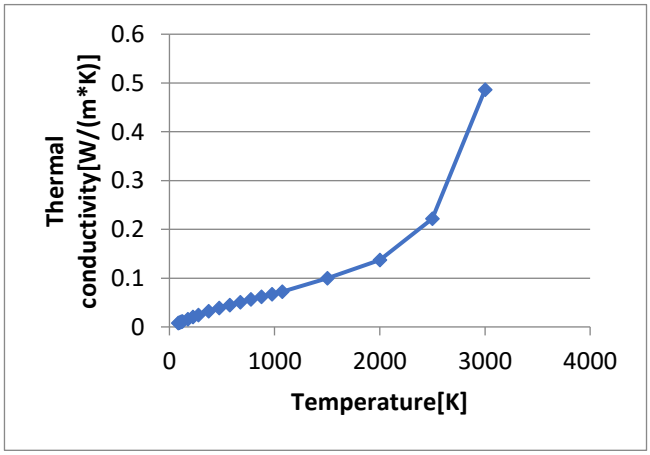
Dynamic viscosity



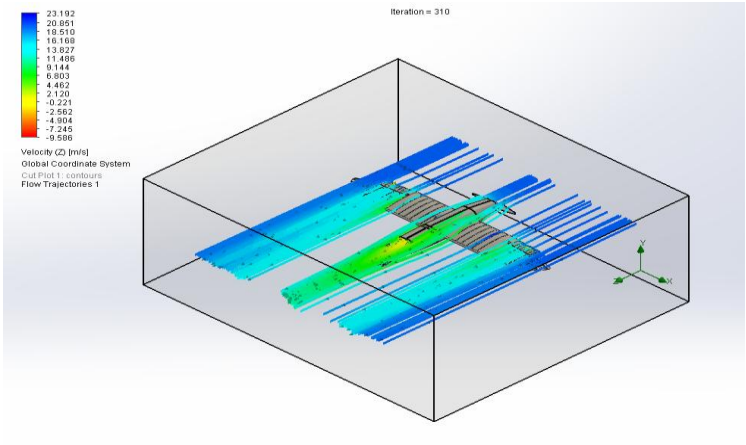
Specific heat (Cp)



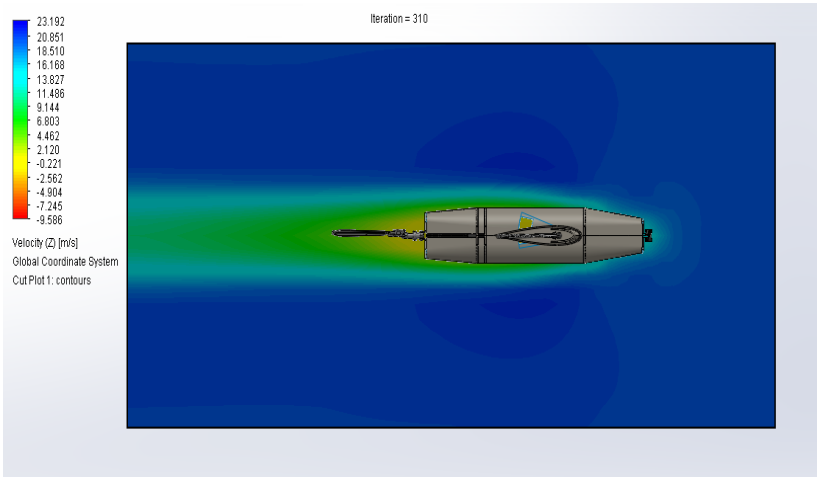
Thermal conductivity



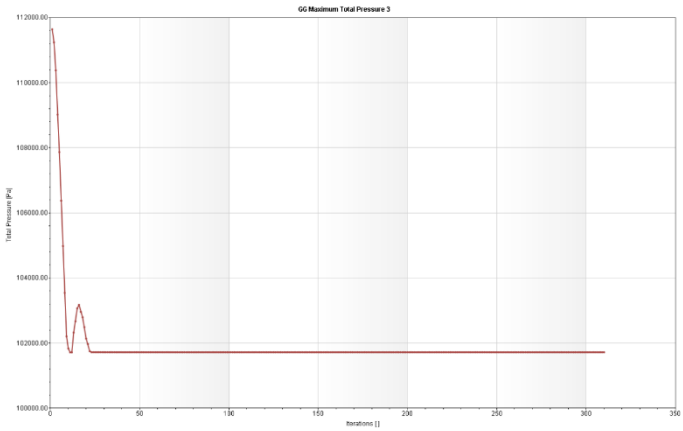
Fluid {Air} Acting on the model {For CRUSING Condition}



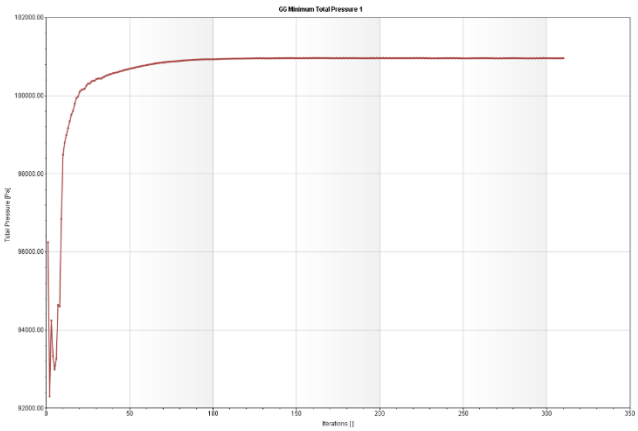
Velocity of air Acting on the Model {For CRUSING Condition}



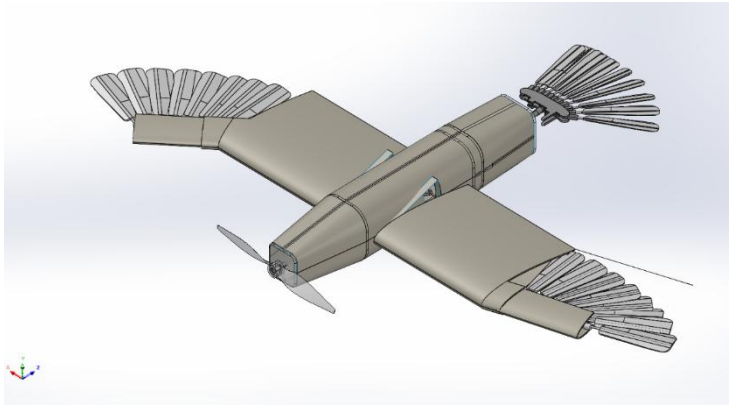
Minimum Total Pressure Graph



Maximum Total Pressure Graph



13.6 Analysis of TURNING Condition.

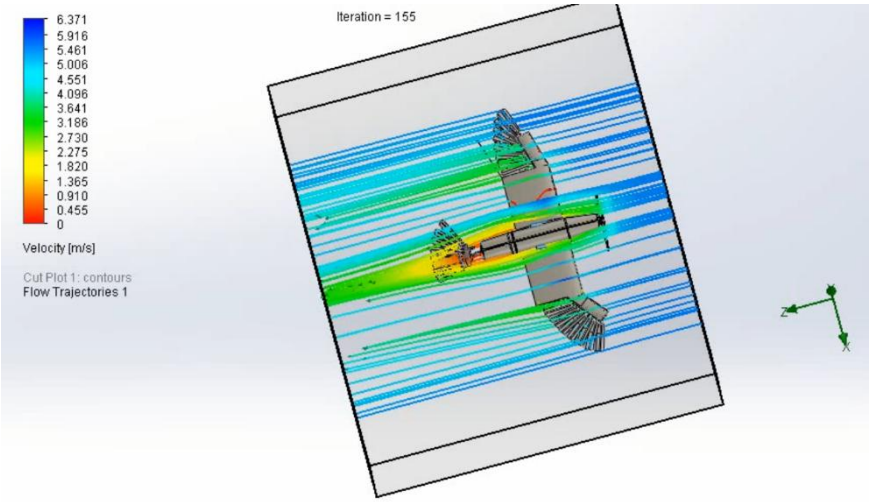


The simulation results indicate a convergent aerodynamic analysis operating at an average flow velocity of approximately 15.90 m/s primarily directed along the Z-axis (fluid flow direction). The simulation captures vital surface and fluid interactions, showing a maximum overall velocity of 18.13 m/s and minimal local backflows or recirculation (indicated by the small negative X and Y localized velocity components). Mechanically, the fluid interaction yields a net Normal/Lift Force (GG Force (Y) 16) of 2.846 N and a localized Friction Force (GG Friction Force (Z) 17) of 0.104 N.

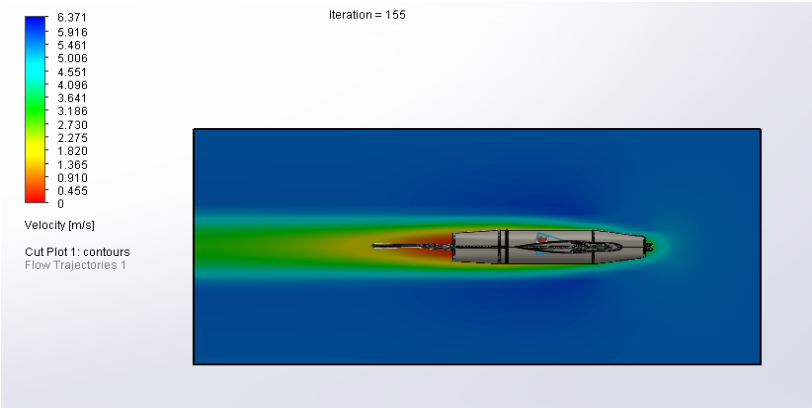
Whether this aircraft will fly depends strictly on its total physical mass:

Yes, it will fly if the total weight of this aircraft model (including its battery, servos, structure, and smart components) is less than 290 grams (equivalent to≈ 2.84 N of gravitational force).

Fluid {Air} Acting on the model {For TURNING Condition}



Velocity of air Acting on the Model {For TURNING Condition}



14. CONCLUSION AND FUTURE WORK

13.1 Conclusion

This study focused on bird-inspired wing morphing and its application in aircraft design. Birds have highly flexible wings that can change shape to achieve efficient flight. Based on this concept, morphing wing technology was studied to improve aircraft performance.

The study shows that morphing wings can increase lift, reduce drag, and improve overall aerodynamic efficiency. It also enhances stability and control under different flight conditions. The use of smart materials and flexible structures plays an important role in achieving morphing.

Overall, bird wing morphing is a promising concept for improving the performance of modern UAVs and future aircraft.

13.2 Future Work

Future work of this project mainly focuses on improving the practical implementation of the morphing wing system. The first step is to develop a physical prototype and test it under controlled conditions such as a wind tunnel to study real aerodynamic behaviour's

Further improvement can be made in the mechanical control system to achieve smoother and more reliable wing morphing. Efficient actuators can be used to increase the speed and accuracy of wing movement.

Detailed study of airflow characteristics and noise reduction can also be carried out, as smoother airflow leads to better performance and quieter operation. The design can be extended to larger aircraft to study structural and performance challenges at a bigger scale.

Finally, real-time testing using a small unmanned aerial vehicle (UAV) can be performed to validate the working of the morphing wing under actual flight conditions.

15. PROJECT SCHEDULE – GANTT CHART

The following Gantt chart shows the complete project timeline from March 9, 2026, to July 4, 2026.

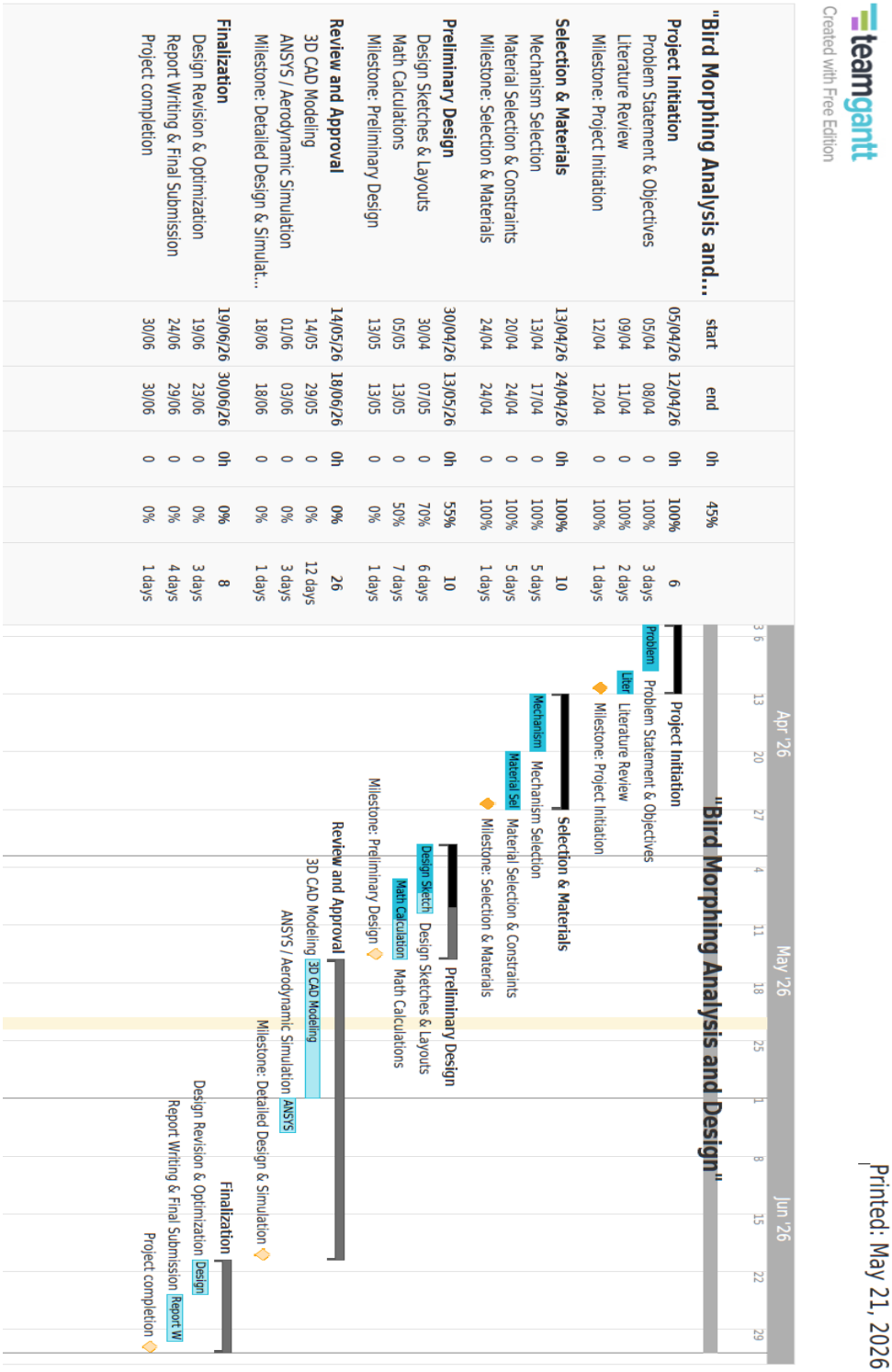


Fig 14.1: Gantt Chart

References

1. Lentin, D. et al. (2007), "How swifts control their glide performance with morphing wings", *Nature*, Vol. 446, No. 7139, pp. 1082–1085.

2. Hartl, D.J. and LA goudas, D.C. (2007), "Aerospace applications of shape memory alloys", *Proceedings of IMechE, Part G: Journal of Aerospace Engineering*, Vol. 221, No. 4, pp. 535–552.

3. Solfa, A.Y.N. et al. (2010), "Shape morphing of aircraft wing: Status and challenges", *Materials and Design*, Vol. 31, No. 3, pp. 1284–1292.

4. Barbarino, S. et al. (2011), "A review of morphing aircraft", *Journal of Intelligent Material Systems and Structures*, Vol. 22, No. 9, pp. 823–877.

5. Vasista, S. et al. (2012), "Realization of morphing wings: A multidisciplinary challenge", *Journal of Aircraft*, Vol. 49, No. 1, pp. 11–28.

6. Ajaj, R.M. et al. (2016), "Morphing aircraft: The need for a new design philosophy", *Aerospace Science and Technology*, Vol. 49, pp. 130–142.

7. Harvey, C. et al. (2019), "Birds can transition between stable and unstable states via wing morphing", *Nature*, Vol. 554, No. 7693, pp. 648–651.

8. Tucker, V.A. (1987), "Gliding birds: The effect of variable wingspan", *Journal of Experimental Biology*, Vol. 133, No. 1, pp. 33–58.

t1.SLDASM [T1 [Normal flight conditions]]

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Minimum Total Pressure 1	[Pa]	101280.1409	101280.0837	101279.8701	101280.3816	100	Yes	0.066180944	42.59946029
GG Average Total Pressure 2	[Pa]	101343.8702	101343.8704	101343.8688	101343.8715	100	Yes	0.000218132	0.010986493
GG Maximum Total Pressure 3	[Pa]	101365.0536	101364.958	101364.8276	101365.0983	100	Yes	0.014311043	31.9450551
GG Minimum Velocity 4	[m/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 5	[m/s]	6.921022499	6.921032828	6.920802696	6.921224405	100	Yes	3.28992E-05	0.00205691
GG Maximum Velocity 6	[m/s]	8.043165195	8.045462245	8.04216609	8.049710557	100	Yes	0.000418822	0.587696033
GG Minimum Velocity (X) 7	[m/s]	-2.22718708	-2.200461739	-2.22973101	-2.172105408	100	Yes	0.003915343	0.145620581
GG Average Velocity (X) 8	[m/s]	-0.00064194	-0.000642987	-0.000684114	-0.000615151	100	Yes	8.79189E-06	2.37658E-05
GG Maximum Velocity (X) 9	[m/s]	1.968217908	1.889661756	1.792840032	1.987614396	100	Yes	0.194774364	0.313587451
GG Minimum Velocity (Y) 10	[m/s]	-2.52489865	-2.529577477	-2.535413147	-2.524279987	100	Yes	0.00045257	0.015924533
GG Average Velocity (Y) 11	[m/s]	0.001186363	0.001235318	0.001122332	0.001319453	100	Yes	3.98943E-05	4.30363E-05
GG Maximum Velocity (Y) 12	[m/s]	2.471280537	2.438163199	2.392712706	2.483661017	100	Yes	0.003347012	0.015405095
GG Minimum Velocity (Z) 13	[m/s]	-6.34500757	-6.422063117	-6.528293204	-6.321889006	100	Yes	0.185201355	0.652071777
GG Average Velocity (Z) 14	[m/s]	6.917796854	6.917829495	6.917580025	6.918010566	100	Yes	3.12755E-05	0.002055678
GG Maximum Velocity (Z) 15	[m/s]	8.009918871	8.014311322	8.00894517	8.022903841	100	Yes	0.000895887	0.033915299
GG Force (Y) 16	[N]	0.013222579	0.013521146	0.011498545	0.015913094	100	Yes	0.00035637	0.003618832
GG Friction Force (Z) 17	[N]	0.033713151	0.033646564	0.033449563	0.033880725	100	Yes	3.5167E-05	0.000487413

Iterations [ ]: 295
Analysis interval: 39

t3 xyz.SLDASM [t3 [LIFT Condition]]

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Minimum Total Pressure 1	[Pa]	101227.8563	101228.3478	101227.4019	101229.7765	100	Yes	0.954573044	95.85948305
GG Average Total Pressure 2	[Pa]	101463.2291	101463.2299	101463.2243	101463.2338	100	Yes	0.002346122	0.034872391
GG Maximum Total Pressure 3	[Pa]	101620.8163	101622.5444	101620.0529	101626.2003	100	Yes	5.822422621	141.7470001
GG Minimum Velocity 4	[m/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 5	[m/s]	15.90755634	15.90763756	15.90713337	15.90797449	100	Yes	0.000268421	0.002480727
GG Maximum Velocity 6	[m/s]	18.04968777	17.99528137	17.92768799	18.07910437	100	Yes	0.071131376	0.524852203
GG Minimum Velocity (X) 7	[m/s]	-5.39599648	-5.514341753	-5.701977144	-5.341200704	100	Yes	0.152132798	0.195445382
GG Average Velocity (X) 8	[m/s]	-6.088E-05	-2.69189E-05	-8.57257E-05	3.93325E-05	51.8	Yes	3.09978E-05	1.6068E-05
GG Maximum Velocity (X) 9	[m/s]	5.561503369	5.554996088	5.468285759	5.62183689	100	Yes	0.063929731	0.224065643
GG Minimum Velocity (Y) 10	[m/s]	-8.60340066	-8.837037382	-9.310456416	-8.588100834	18.3	Yes	0.248540502	0.045669777
GG Average Velocity (Y) 11	[m/s]	-0.06001403	-0.060032732	-0.06026695	-0.059693084	100	Yes	0.000386132	0.001789494
GG Maximum Velocity (Y) 12	[m/s]	3.816996717	3.710747879	3.368170001	4.291196813	36.7	Yes	0.108806585	0.04002855
GG Minimum Velocity (Z) 13	[m/s]	-5.50273253	-5.699108807	-5.906899941	-5.494970376	100	Yes	0.039242333	0.894100563
GG Average Velocity (Z) 14	[m/s]	15.90261659	15.90259048	15.90218976	15.90282571	100	Yes	0.000206766	0.002523429
GG Maximum Velocity (Z) 15	[m/s]	18.04751305	17.98930573	17.91881415	18.0766875	26.8	Yes	0.068737558	0.018457754
GG Force (Y) 16	[N]	3.153456679	3.162389184	3.150795867	3.171494002	100	Yes	0.015616232	0.15520379
GG Friction Force (Z) 17	[N]	0.0895307	0.089905985	0.087579437	0.091834709	100	Yes	0.000337462	0.002755378

Iterations [ ]: 264
Analysis interval: 33